# Application Note
# Register Maps

## ABSTRACT

RAM registers mainly contain system parameters, calibration, pin remapping (NTC, GPIO, etc.), primary and secondary protection parameters, as well as balancing registers. Sysc refers to system registers, which include operating modes, communication methods, LDO output, MOSFET control, etc. Pad is related to the parameter settings for multi-function pin multiplexing, with a base address of 0x8200. Adc pertains to calibration-related parameter settings, most of which have already been calibrated at the factory. Protect includes all primary and secondary protection-related parameter settings. CB refers to parameter settings related to balancing and load detection.

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# 1 Sys

Base Address: 0x8000

## 1.1 Power Config Register

Address Offset: 0x00

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| normal_intr_clr | [15:15] | 0x0 | RW1P | write 1 to clr normal intr |
| normal_intr_mask | [14:14] | 0x0 | RW | 0: enable normal intr; 1: mask normal intr，enable alarm assert |
| dis_sleep2normal | [13:13] | 0x0 | RW | 0: not disable sleep mode jump to normal mode<br>1: disable sleep mode jump to normal mode |
| shut_ts2 | [12:12] | 0x0 | RW | When TS2 is used as a thermistor or has some other external pulldown, SHUTDOWN mode cannot be used. If TS2 is externally pulled down, this bit should be set to prevent the device from immediately exiting SHUTDOWN mode in some circumstances. When this bit is set and SHUTDOWN conditions are met, the device will instead enter a low power state waiting for a rising edge on the LD pin.<br>0 = Standard SHUTDOWN mode behavior is selected<br>1 = SHUTDOWN mode replaced by low-power state waiting for rising edge on LD pin |
| dpslp_pd | [11:11] | 0x1 | RW | Enables wake from DEEPSLEEP based on charger attach. When clear, DEEPSLEEP mode must be exited via host command.<br>0 = DEEPSLEEP not exited on rising edge on LD pin<br>1 = A rising edge on the LD pin exits DEEPSLEEP |
| dpslp_ldo | [10:10] | 0x1 | RW | Determines whether or not REG1 and REG2 are disabled in DEEPSLEEP mode<br>0 = Disable REG1 and REG2 when entering DEEPSLEEP mode 1 = Leave REG1 and REG2 in present state when entering DEEPSLEEP |
| dpslp_lfo | [ 9: 9] | 0x1 | RW | Determines whether or not to disable the Low Frequency Oscillator in DEEPSLEEP mode to conserve power.<br>0 = Disable the Low Frequency Oscillator in DEEPSLEEP mode (recommended)<br>1 = Enable the Low Frequency Oscillator in DEEPSLEEP mode |
| sleep | [ 8: 8] | 0x1 | ROHW | Sets the default value of BatteryStatus()[SLEEP_EN] which enables or disables SLEEP mode. After initialization, SLEEP_EN can stil be changed via the SLEEP_ENABLE and SLEEP_DISABLE subcommands. |


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| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| | | | | 0 = Disable SLEEP mode by default<br>1 = Enable SLEEP mode by default |
| otsd | [ 7: 7] | 0x1 | RW | Enables or disables the on-chip over-temperature detection circuit to shut down the device in case of a severe on-chip over-temperature condition.<br>0 = Disable SHUTDOWN from on-chip over-temperature detection circuit (not recommended)<br>1 = Enter SHUTDOWN mode when on-chip over-temperature condition is detected |
| dpslp_icg_en | [ 6: 6] | 0x0 | RW | 0: deepsleep mode,icg disable; 1: deepsleep mode,icg enable |
| mcu_wdg_tim | [ 5: 4] | 0x0 | RW | mcu wdg timeout |
| loop_slow | [ 3: 2] | 0x0 | RW | Selects normal ADC scan loop speed by inserting current-only measurements after each voltage and temperature scan loop. This setting is used while cell balancing is not active.<br>0 = Full speed<br>1 = Half speed<br>2 = Quarter speed<br>3 = Eighth speed |
| wk_spd | [ 1: 0] | 0x2 | RW | Selects coulomb counter conversion speed used in SLEEP mode for the current wake comparator function. Slower conversion speed results in lower noise.<br>Setting 0x2 may exhibit a large offset and should be avoided. Setting 0x3 results in ~100µV (1-sigma) noise level, so should only be used when the Wake Comparator Current is set to a level such that \|VSRP - VSRN\| > 1000µV.<br>0x0 = 48 ms conversion rate<br>0x1 = 24 ms conversion rate<br>0x2 = 12 ms conversion rate<br>0x3 = 6 ms conversion rate |

## 1.2 REG12 Config Register

Address Offset: 0x02

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| reg2v | [ 7: 5] | 0x0 | RWHW | Selects voltage level for REG2 This setting should not be changed while REG2 is enabled.<br>0 = 3.3V<br>1 = 5V<br>2 = 3V |


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| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| | | | | 3 = 2.5 V<br>4-7 = 1.8V |
| reg2_en | [ 4: 4] | 0x0 | RWHW | Configure default state for REG2 output. This setting is reapplied when initializing after reset or DEEPSLEEP mode.<br>0 = REG2 Disabled<br>1 = REG2 Enabled |
| reg1v | [ 3: 1] | 0x0 | RWHW | Selects voltage level for REG1 This setting should not be changed while REG1 is enabled.<br>0 = 3.3V<br>1 = 5V<br>2 = 3V<br>3 = 2.5 V<br>4-7 = 1.8V |
| reg1_en | [ 0: 0] | 0x1 | RWHW | Configure default state for REG1 output. This setting is reapplied when initializing after reset or DEEPSLEEP mode.<br>0 = REG1 Disabled<br>1 = REG1 Enabled |

## 1.3 REG0 Config Register

Address Offset: 0x03

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| reg0_en | [ 0: 0] | 0x1 | RW | Enable or disable pre-regulator for REG1 and REG2<br>0 = Pre-regulator is disabled<br>1 = Pre-regulator is enabled |

## 1.4 HWD Regulator Options Register

Address Offset: 0x04

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| toggle_opt | [ 5: 4] | 0x0 | RW | Action to take when HWD protection is triggered 0 = Take no action on REG1 and REG2 regulators<br>1 = Turn REG1 and REG2 regulators off<br>2 = Turn REG1 and REG2 regulators off for configured duration and then back on again<br>3 = Reserved. Do not use |
| toggle_time | [ 3: 0] | 0x0 | RW | How long to keep REG1 and REG2 regulators off when configured to toggle<br>0 = Turn REG1 and REG2 regulators off and do not turn back on again<br>1–15 = Turn REG1 and REG2 regulators back on again |


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| | | | | |
|---|---|---|---|---|
| | | | | after this many seconds |

## 1.5 Comm Type Register

Address Offset: 0x05

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| comm_type | [ 7: 0] | 0x0 | RW | 0x00 = Default 0xff = Default<br>0x07 = I2C (for use up to 100 kHz bus speed)<br>0x08 = I2C Fast (for use above 100 kHz bus speed)<br>0x09 = I2C Fast with Timeouts (for use above 100 kHz bus speed)<br>0x0f = SPI<br>0x10 = SPI with CRC<br>0x11 = I2C with CRC (for use up to 100 kHz bus speed)<br>0x12 = I2C Fast with CRC (for use above 100 kHz bus speed)<br>0x1e = I2C with Timeouts (for use up to 100 kHz bus speed)<br>All other values = Reserved. Do not use |

## 1.6 I2C Address Register

Address Offset: 0x06

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| i2c_addr | [ 6: 0] | 0x10 | RW | Sets the device address for I2C-based communication modes. This applied on reset or when SWAP_COMM_MODE subcommand is received. |

## 1.7 SPI Configuration Register

Address Offset: 0x07

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| miso_reg1 | [ 5: 5] | 0x0 | RW | Configures SPI MISO pin to use REG1 output drive level |
| filt | [ 4: 4] | 0x0 | RW | Configures digital filters on SPI input pins<br>0 = Disable digital filters on SPI input pins (recommended for high-frequency operation)<br>1 = Enable digital filters on SPI input pins |

## 1.8 Comm Idle Time Register

Address Offset: 0x08

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|


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| | | | | |
|---|---|---|---|---|
| com_idletime | [ 7: 0] | 0x0 | RW | 0 = High-Frequency Oscillator remains enabled after communication ends.<br>All other values = High-Frequency Oscillator left enabled for up to this many seconds after communications.<br>After closing, to re-establish communication, it is necessary to first send a wake-up message to initiate communication with the AFE. |

## 1.9 Low Shutdown Delay Register

Address Offset: 0x09

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| lowv_shutdown_dly | [ 7: 0] | 0x1 | RW | Configures the number of seconds the device must measure voltage below Power: Shutdown: Shutdown Stack Voltage or Power: Shutdown: Shutdown Cell Voltage to enter SHUTDOWN mode. A value of 0 means that shutdown will be entered with a single low voltage sample. The unit is S. |

## 1.10 Shutdown Cell Voltage Register

Address Offset: 0x0A

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| shutdown_cell_volt | [15: 0] | 0x0 | RW | Configures the cell voltage threshold at which the device will enter SHUTDOWN mode. This threshold does not apply to VC pins configured as interconnects.<br>0 = Cell-Voltage-based shutdown disabled<br>All other values = Cell voltage shutdown threshold<br>The unit is mV. |

## 1.11 Shutdown Stack Voltage Register

Address Offset: 0x0C

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| shutdown_stack_volt | [15: 0] | 0x258 | RW | Configures the pack voltage threshold at which the device will enter SHUTDOWN mode. The unit is userV, default is 8mV. |

## 1.12 Shutdown Temp Register

Address Offset: 0x0E

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| shutdown_temp | [15: 0] | 0xdfd | RW | Configures the internal temperature threshold at which the device will shut down.<br>0 = Shutdown based on measured internal temperature disabled All other values = Shutdown Internal Temperature threshold. |


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| | | | | |
|---|---|---|---|---|
| | | | | The unit of temperature is 0.1℃. |

## 1.13 Shutdown Temperature Delay Register

Address Offset: 0x10

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| shutdown_temp_dly | [ 7: 0] | 0x5 | RW | Configures the number of consecutive seconds the device must measure an internal temperature above Power: Shutdown: Shutdown Temperature to enter SHUTDOWN mode. |

## 1.14 Shutdown Command Delay Register

Address Offset: 0x11

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| shutdown_cmd_dly | [ 7: 0] | 0x0 | RW | When the SHUTDOWN command is received or the RST_SHUT pin is detected high for one second, the device will enter SHUTDOWN after this delay. If the LD pin voltage is still high after this delay, the device will delay entering SHUTDOWN further until that voltage is no longer present. |

## 1.15 Auto Shutdown Time Register

Address Offset: 0x12

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| auto_shutdown_tim | [ 7: 0] | 0x0 | RW | As a countermeasure to inadvertent wake from SHUTDOWN, the device can be configured to automatically enter SHUTDOWN again after a number of seconds defined by this parameter. If communications occur after the first second, or if charge current is detected, or if discharge current above the SLEEP current threshold is detected after one second, automatic shutdown is cancelled. If none of those events occur, after this number of seconds, the FETs will be turned off for up to five seconds. During that 5 seconds, if the PACK and LD pin voltages fall, the device will enter shutdown. If the voltage on those pins remain above their respective thresholds, auto-shutdown is cancelled based on a conclusion that a charger is present.<br>0 = Auto-shutdown feature is disabled.<br>All other values = Auto-shutdown occurs after this many seconds if it is not cancelled. |


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## 1.16 Manufacturing Register

Address Offset: 0x13

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| otpw_en | [ 7: 7] | 0x0 | RW | This bit enables or disables writes to OTP during operation. The device can program diagnostic information in OTP when a PF occurs or when the fuse is blown to retain state after a full reset. It also can program MANU_DATA upon request (if in FULLACCESS mode). This bit enables the device to program this runtime data to OTP. Programming will only occur when the stack voltage and temperature are within allowed limits. If this bit is not set, programming may still be done in CONFIG_UPDATE mode.<br>0 = Device will not program OTP during operation.<br>1 = Device may program OTP during operation. |
| pf_en | [ 6: 6] | 0x1 | RWHW | This bit enables or disables Permanent Failure checks. Clearing this bit prevents Permanent Failure from triggering which is useful during manufacturing.<br>0 = Permanent Failure checks are disabled.<br>1 = Permanent Failure checks are enabled |
| fet_en | [ 4: 4] | 0x0 | RWHW | This bit enables or disables FET Test mode. In FET Test mode, the FET states are controlled by the FET Test subcommands. This is typically used during manufacturing to test FET circuitry. Note that safety checks still may force FETs off (or for body diode protection, on) in FET Test mode.<br>0 = Normal FET control is disabled. FET Test mode is enabled. Device will not turn on FETs unless FET Test subcommands instruct it to do so.<br>1 = Normal FET control is enabled. FET Test mode is disabled. Device will ignore FET Test subcommands. |

## 1.17 Sleep Current Register

Address Offset: 0x14

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| sleep_curr | [15: 0] | 0x32 | RW | Configures the current threshold above which the device will not enter SLEEP mode. If current is measured above this value during a periodic measurement in SLEEP mode, SLEEP mode will be exited. The unit is userA. UserA is user-defined in cc gain . |


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## 1.18 Sleep Hysteresis Register

Address Offset: 0x16

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| sleep_hyst_tim | [ 7: 0] | 0xa | RW | After transitioning to NORMAL mode, the device will not enter SLEEP mode again until this number of seconds has passed. |

## 1.19 Voltage Time Register

Address Offset: 0x17

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| volt_tim | [ 7: 0] | 0x5 | RW | This parameter sets how often the device wakes to measure voltages and temperatures while in SLEEP mode. While in sleep mode, the device begins 4-second current measurements one second after measuring voltages and temperatures. These measurements interrupt the 4-second current measurement and start it over. That means that setting this parameter to a value less than 5 seconds will result in no completed 4second current measurements in sleep. The most common settings of this parameter are 1 second, 5 seconds, or (4 * n + 1) seconds. |

## 1.20 Wake Comparator Current Register

Address Offset: 0x18

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| wake_comp_curr | [15: 0] | 0x1f4 | RW | Configures the current wake comparator threshold in sleep mode. Note that the default wake comparator conversion speed (12 ms) results in lower accuracy conversions than the normal current measurements. The smallest recommended threshold at this speed is 500userA. |

## 1.21 Sleep Charger Voltage Threshold Register

Address Offset: 0x1A

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| sleep_chg_volt_thold | [15: 0] | 0x7d0 | RW | The Top-of-Stack voltage must be below this threshold to block SLEEP mode based on charger presence. The unit is userV, default is 8mV. |

## 1.22 Sleep Charger Packtos Delta Register

Address Offset: 0x1C

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|


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| | | | | |
|---|---|---|---|---|
| sleep_chg_pack_delta | [15: 0] | 0xc8 | RW | If the PACK pin voltage is higher than the Top-of-Stack by more than this threshold and the Top-of Stack voltage is less than Power: Sleep: Sleep Charger Voltage Threshold, SLEEP mode is not allowed. This threshold is also used for charger detection when delaying voltage-based shutdown. The unit is userV, default is 8mV. |

## 1.23 Cell1 Interconnect Register

Address Offset: 0x1E

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cell1_intercon | [15: 0] | 0x0 | RW | When a significant interconnect resistance exists between cells, this parameter can be used to account for it in cell voltage measurements. Cell voltage calculations utilize the simultaneous current measurement and this resistance value to accurately calculate the cell voltage. |

## 1.24 Cell2 Interconnect Register

Address Offset: 0x20

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cell2_intercon | [15: 0] | 0x0 | RW | When a significant interconnect resistance exists between cells, this parameter can be used to account for it in cell voltage measurements. Cell voltage calculations utilize the simultaneous current measurement and this resistance value to accurately calculate the cell voltage. |

## 1.25 Cell3 Interconnect Register

Address Offset: 0x22

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cell3_intercon | [15: 0] | 0x0 | RW | When a significant interconnect resistance exists between cells, this parameter can be used to account for it in cell voltage measurements. Cell voltage calculations utilize the simultaneous current measurement and this resistance value to accurately calculate the cell voltage. |

## 1.26 Cell4 Interconnect Register

Address Offset: 0x24

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cell4_intercon | [15: 0] | 0x0 | RW | When a significant interconnect resistance exists between cells, this parameter can be used to account for it in cell voltage measurements. Cell voltage calculations utilize the simultaneous current measurement and this resistance value to accurately calculate the cell voltage. |


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## 1.27 Cell5 Interconnect Register

Address Offset: 0x26

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cell5_intercon | [15: 0] | 0x0 | RW | When a significant interconnect resistance exists between cells, this parameter can be used to account for it in cell voltage measurements. Cell voltage calculations utilize the simultaneous current measurement and this resistance value to accurately calculate the cell voltage. |

## 1.28 Cell6 Interconnect Register

Address Offset: 0x28

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cell6_intercon | [15: 0] | 0x0 | RW | When a significant interconnect resistance exists between cells, this parameter can be used to account for it in cell voltage measurements. Cell voltage calculations utilize the simultaneous current measurement and this resistance value to accurately calculate the cell voltage. |

## 1.29 Cell7 Interconnect Register

Address Offset: 0x2A

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cell7_intercon | [15: 0] | 0x0 | RW | When a significant interconnect resistance exists between cells, this parameter can be used to account for it in cell voltage measurements. Cell voltage calculations utilize the simultaneous current measurement and this resistance value to accurately calculate the cell voltage. |

## 1.30 Cell8 Interconnect Register

Address Offset: 0x2C

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cell8_intercon | [15: 0] | 0x0 | RW | When a significant interconnect resistance exists between cells, this parameter can be used to account for it in cell voltage measurements. Cell voltage calculations utilize the simultaneous current measurement and this resistance value to accurately calculate the cell voltage. |


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## 1.31 Cell9 Interconnect Register

Address Offset: 0x2E

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cell9_intercon | [15: 0] | 0x0 | RW | When a significant interconnect resistance exists between cells, this parameter can be used to account for it in cell voltage measurements. Cell voltage calculations utilize the simultaneous current measurement and this resistance value to accurately calculate the cell voltage. |

## 1.32 Reserve Register

Address Offset: 0x30

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| Reserve | [15: 0] | 0x0 | RW | When a significant interconnect resistance exists between cells, this parameter can be used to account for it in cell voltage measurements. Cell voltage calculations utilize the simultaneous current measurement and this resistance value to accurately calculate the cell voltage. |

## 1.33 Cell10 Interconnect Register

Address Offset: 0x32

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cell10_intercon | [15: 0] | 0x0 | RW | When a significant interconnect resistance exists between cells, this parameter can be used to account for it in cell voltage measurements. Cell voltage calculations utilize the simultaneous current measurement and this resistance value to accurately calculate the cell voltage. |

## 1.34 Cell11 Interconnect Register

Address Offset: 0x34

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cell11_intercon | [15: 0] | 0x0 | RW | When a significant interconnect resistance exists between cells, this parameter can be used to account for it in cell voltage measurements. Cell voltage calculations utilize the simultaneous current measurement and this resistance value to accurately calculate the cell voltage. |


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## 1.35 Cell12 Interconnect Register

Address Offset: 0x36

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cell12_intercon | [15: 0] | 0x0 | RW | When a significant interconnect resistance exists between cells, this parameter can be used to account for it in cell voltage measurements. Cell voltage calculations utilize the simultaneous current measurement and this resistance value to accurately calculate the cell voltage. |

## 1.36 Cell13 Interconnect Register

Address Offset: 0x38

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cell13_intercon | [15: 0] | 0x0 | RW | When a significant interconnect resistance exists between cells, this parameter can be used to account for it in cell voltage measurements. Cell voltage calculations utilize the simultaneous current measurement and this resistance value to accurately calculate the cell voltage. |

## 1.37 Cell14 Interconnect Register

Address Offset: 0x3A

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cell14_intercon | [15: 0] | 0x0 | RW | When a significant interconnect resistance exists between cells, this parameter can be used to account for it in cell voltage measurements. Cell voltage calculations utilize the simultaneous current measurement and this resistance value to accurately calculate the cell voltage. |

## 1.38 Cell15 Interconnect Register

Address Offset: 0x3C

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cell15_intercon | [15: 0] | 0x0 | RW | When a significant interconnect resistance exists between cells, this parameter can be used to account for it in cell voltage measurements. Cell voltage calculations utilize the simultaneous current measurement and this resistance value to accurately calculate the cell voltage. |


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## 1.39 Cell16 Interconnect Register

Address Offset: 0x3E

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cell16_intercon | [15: 0] | 0x0 | RW | When a significant interconnect resistance exists between cells, this parameter can be used to account for it in cell voltage measurements. Cell voltage calculations utilize the simultaneous current measurement and this resistance value to accurately calculate the cell voltage. |

## 1.40 Unseal Keystep1 Register

Address Offset: 0x40

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| unseal_key_step1 | [15: 0] | 0x5a5a | RWHW | This is the first word of the security key that must be sent to transition from SEALED to UNSEALED mode. |

## 1.41 Unseal Keystep2 Register

Address Offset: 0x42

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| unseal_key_step2 | [15: 0] | 0x5a5a | RWHW | This is the second word of the security key that must be sent to transition from SEALED to UNSEALED mode. It must be sent within 5 seconds of the first word of the key and with no other commands in between. |

## 1.42 Full Access Keystep1 Register

Address Offset: 0x44

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| full_acc_key_step1 | [15: 0] | 0xa5a5 | RWHW | This is the second word of the security key that must be sent to transition from UNSEALED to FULL ACCESS mode. |

## 1.43 Full Access Keystep2 Register

Address Offset: 0x46

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| full_acc_key_step2 | [15: 0] | 0xa5a5 | RWHW | This is the second word of the security key that must be sent to transition from UNSEALED to FULL ACCESS mode. It must be sent within 5 seconds of the first word of the key and with no other commands in between. |


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## 1.44 Security Settings Register

Address Offset: 0x48

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| perm_seal | [ 2: 2] | 0x0 | RW | Setting this bit prevents unsealing the device once it is sealed. If this is not programmed to OTP, this setting will be lost on a full reset and the device will again be able to UNSEAL.<br>0 = The device can be unsealed by sending the correct security keys.<br>1 = The device cannot be unsealed. |
| lock_cfg | [ 1: 1] | 0x0 | RW | Setting this bit prevents entry into CONFIG_UPDATE and FULLACCESS modes. This prevents further modifications to the device configuration after CONFIG_UPDATE mode is exited.<br>0 = Configuration parameters can be changed in CONFIG_UPDATE mode.<br>1 = Configuration parameters cannot be changed. |
| seal | [ 0: 0] | 0x0 | RW | Setting this bit causes the device to enter SEALED mode when reset or exiting CONFIG_UPDATE and DEEPSLEEP modes. In production systems, this bit should be set for security purposes.<br>0 = Device does not default to SEALED mode.<br>1 = Device default state is SEALED. |

## 1.45 ADC Clk Cfg Register

Address Offset: 0x49

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cadc_clk_sel1 | [ 7: 6] | 0x0 | RW | Sleep/DeepSleep mode：curr adc clk divider<br>0: 1 Frequency division<br>1: 2 Frequency division<br>2: 4 Frequency division<br>3: 8 Frequency division |
| cadc_clk_sel0 | [ 5: 4] | 0x0 | RW | Normal mode：curr adc clk divider<br>0: 1 Frequency division<br>1: 2 Frequency division<br>2: 8 Frequency division<br>3: 24 Frequency division |
| vadc_clk_sel1 | [ 3: 2] | 0x0 | RW | Sleep/DeepSleep mode：volt adc clk divider<br>0: 1 Frequency division<br>1: 2 Frequency division<br>2: 4 Frequency division<br>3: reserve writing '3' is prohibited |


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| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| vadc_clk_sel0 | [ 1: 0] | 0x0 | RW | Normal mode：volt adc clk divider<br>0: 1 Frequency division<br>1: 2 Frequency division<br>2: 4 Frequency division<br>3: reserve writing '3' is prohibited |

## 1.46 OTP Cfg Register

Address Offset: 0x4A

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| otp_pgm_mode | [14:14] | 0x0 | RW | otp pgm mode<br>0 = continue pgm<br>1 = single pgm |
| otp_pgm_en | [13:13] | 0x0 | RW | otp pgm enable |
| otp_test_mode | [12: 8] | 0x0 | RW | otp test mode |
| otp_dsb_en | [ 7: 7] | 0x0 | RW | otp deep standby mode |
| otp_clk_en | [ 6: 6] | 0x1 | RW | otp clk enable |
| otp_test_margin | [ 5: 0] | 0x0 | RW | otp test margin |

## 1.47 WDG EN Register

Address Offset: 0x4C

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| mcu_wdg_en | [ 1: 1] | 0x0 | RW | AFE communication watchdog enable |
| adc_wdg_en | [ 0: 0] | 0x0 | RW | adc wdg enable |

## 1.48 FET Off Delay Register

Address Offset: 0x4D

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| fet_off_dly | [ 7: 0] | 0x0 | RW | When the SHUTDOWN command is received or the RST_SHUT pin is detected high for one second, the FETs are turned off after this delay. |

## 1.49 Scan Mode Register

Address Offset: 0x4E

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| scan_mode_support | [ 7: 7] | 0x0 | RWONCE | 0 = not support<br>1 = support use register to enter scan mode; |
| scan_mode | [ 0: 0] | 0x0 | RWONCE | 1 = scan mode |


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# 2 Pad

Base Address: 0x8200

## 2.1 CFETOFF Pin Config Register

Address Offset: 0x00

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cfetoff_opt | [ 7: 2] | 0x4 | RW | The options for pin functions selected by these bit configurations are as follows:<br>OPT[5]: Polarity when configured as CFETOFF (no effect on GPO mode)<br>0 = Select active high<br>1 = Select active low;<br>OPT[4]:<br>0 = Digital output (OEN=0)<br>1 = No digital output (OEN=1, HZ)<br>OPT[3]:<br>0 = Output high-level drive using REG1<br>1 = Output high-level drive using REG18<br>OPT[2]:<br>0 = Disable digital input (IE=0)<br>1 = Enable digital input (IE=1)<br>OPT[1]:<br>0 = Disable digital input weak pull-up<br>1 = Enable digital input weak pull-up<br>OPT[0]:<br>0 = Disable digital input weak pull-down<br>1 = Enable digital input weak pull-down<br><br>When selecting pins for ADCIN or thermistor functions, the OPT[5:0] bits will be used as follows:<br>OPT[5:4]: If unused, set OPT[4] to 1.<br>OPT[3]: Whether the pin is used as ADC input:<br>0 = No<br>1 = Yes<br>OPT[2]: Whether the pin is used as Tsensor input with pull-up resistor<br>0 = No<br>1 = Yes<br>OPT[1]:<br>0 = Not used for FET temperature protection<br>1 = Used for FET temperature protection<br>OPT[0]: |


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| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| | | | | 0 = Not used for cell temperature protection<br>1 = Used for cell temperature protection |
| cfetoff_pin_fxn | [ 1: 0] | 0x0 | RW | cfetoff_pin_fxn [1:0] 0x0 RW These bits configure which function this pin is used for:<br>0 = SPI_CS<br>1 = General Output<br>2 = CFETOFF<br>3 = ADC Input or thermistor |

## 2.2 DFETOFF Pin Config Register

Address Offset: 0x01

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| dfetoff_opt | [ 7: 2] | 0x0 | RW | The options for pin functions selected by these bit configurations are as follows:<br>OPT[5]:<br>Polarity when configured as DFETOFF (no effect on GPO mode)<br>0 = Select active high<br>1 = Select active low<br>OPT[4]:<br>0 = Digital output (OEN=0)<br>1 = No digital output (OEN=1, HZ)<br>OPT[3]:<br>0 = Output high-level drive using REG1<br>1 = Output high-level drive using REG18<br>OPT[2]:<br>0 = Disable digital input (IE=0)<br>1 = Enable digital input (IE=1)<br>OPT[1]:<br>0 = Disable digital input weak pull-up<br>1 = Enable digital input weak pull-up<br>OPT[0]:<br>0 = Disable digital input weak pull-down<br>1 = Enable digital input weak pull-down<br><br>When selecting pins for ADCIN or thermistor functions, the OPT[5:0] bits will be used as follows:<br>OPT[5:4]: If unused, set OPT[4] to 1.<br>OPT[3]: Whether the pin is used as ADC input:<br>0 = No<br>1 = Yes<br>OPT[2]: Whether the pin is used as Tsensor input with pull- |


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| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| | | | | up resistor:<br>0 = No<br>1 = Yes<br>OPT[1]:<br>0 = Not used for FET temperature protection<br>1 = Used for FET temperature protection<br>OPT[0]:<br>0 = Not used for cell temperature protection<br>1 = Used for cell temperature protection |
| dfetoff_pin_fxn | [ 1: 0] | 0x0 | RW | These bits configure which function this pin is used for.<br>0 = DFETOFF<br>1 = General Purpose Output<br>2 = BOTHOFF<br>3 = ADC Input or Thermistor |

## 2.3 Alert Pin Config Register

Address Offset: 0x02

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| alert_opt | [ 7: 2] | 0x0 | RW | The options for pin functions selected by these bit configurations are as follows:<br>OPT[5]:<br>Polarity when configured as ALERT (no effect on GPO mode)<br>0 = Select active high<br>1 = Select active low<br>OPT[4]:<br>0 = Digital output (OEN=0)<br>1 = No digital output (OEN=1, HZ)<br>OPT[3]:<br>0 = Output high-level drive using REG1<br>1 = Output high-level drive using REG18<br>OPT[2]:<br>0 = Disable digital input (IE=0)<br>1 = Enable digital input (IE=1)<br>OPT[1]:<br>0 = Disable digital input weak pull-up<br>1 = Enable digital input weak pull-up<br>OPT[0]:<br>0 = Disable digital input weak pull-down<br>1 = Enable digital input weak pull-down<br><br>When selecting pins for ADCIN or thermistor functions, the |


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| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| | | | | OPT[5:0] bits will be used as follows:<br>OPT[5:4]: If unused, set OPT[4] to 1.<br>OPT[3]: Whether the pin is used as ADC input:<br>0 = No<br>1 = Yes<br>OPT[2]: Whether the pin is used as Tsensor input with pull-up resistor:<br>0 = No<br>1 = Yes<br>OPT[1]:<br>0 = Not used for FET temperature protection<br>1 = Used for FET temperature protection<br>OPT[0]:<br>0 = Not used for cell temperature protection<br>1 = Used for cell temperature protection |
| alert_pin_fxn | [ 1: 0] | 0x0 | RW | These bits configure which function this pin is used for:<br>0 = Unused<br>1 = General Output<br>2 = ALERT<br>3 = ADC Input or Thermistor |

## 2.4 Reserved Pin Config Register

Address Offset: 0x03

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| Reserved | [ 7: 0] | 0x0 | RW | Unused |

## 2.5 SCL Pin Config Register

Address Offset: 0x04

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| scl_opt | [ 7: 2] | 0x4 | RW | The options for pin functions selected by these bit configurations are as follows:<br>OPT[5]: Not used<br>OPT[4]:<br>0 = Digital output (OEN=0)<br>1 = No digital output (OEN=1, HZ)<br>OPT[3]:<br>0 = Output high-level drive using REG1<br>1 = Output high-level drive using REG18<br>OPT[2]: |


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| | | | | |
|---|---|---|---|---|
| | | | | 0 = Disable digital input (IE=0)<br>1 = Enable digital input (IE=1)<br>OPT[1]:<br>0 = Disable digital input weak pull-up<br>1 = Enable digital input weak pull-up<br>OPT[0]:<br>0 = Disable digital input weak pull-down<br>1 = Enable digital input weak pull-down |
| scl_pin_fxn | [ 1: 0] | 0x0 | RW | These bits configure which function this pin is used for:<br>0 = SCL or SPI_SCLK<br>1 = Not used<br>2 = Not used<br>3 = Not used |

## 2.6 SDA Pin Config Register

Address Offset: 0x05

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| sda_opt | [ 7: 2] | 0x4 | RW | The options for pin functions selected by these bit configurations are as follows:<br>OPT[5]: Not used<br>OPT[4]:<br>0 = Digital output (OEN=0)<br>1 = No digital output (OEN=1, HZ)<br>OPT[3]:<br>0 = Output high-level drive using REG1<br>1 = Output high-level drive using REG18<br>OPT[2]:<br>0 = Disable digital input (IE=0)<br>1 = Enable digital input (IE=1)<br>OPT[1]:<br>0 = Disable digital input weak pull-up<br>1 = Enable digital input weak pull-up<br>OPT[0]:<br>0 = Disable digital input weak pull-down<br>1 = Enable digital input weak pull-down |
| sda_pin_fxn | [ 1: 0] | 0x0 | RW | These bits configure which function this pin is used for:<br>0 = SDA or SPI_MISO<br>1 = Not used<br>2 = Not used<br>3 = Not used |


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## 2.7 SPI MOSI Pin Config Register

Address Offset: 0x06

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| spi_mosi_opt | [ 7: 2] | 0x5 | RW | The options for pin functions selected by these bit configurations are as follows:<br>OPT[5]: Not used<br>OPT[4]:<br>0 = Digital output (OEN=0)<br>1 = No digital output (OEN=1, HZ)<br>OPT[3]:<br>0 = Output high-level drive using REG1<br>1 = Output high-level drive using REG18<br>OPT[2]:<br>0 = Disable digital input (IE=0)<br>1 = Enable digital input (IE=1)<br>OPT[1]:<br>0 = Disable digital input weak pull-up<br>1 = Enable digital input weak pull-up<br>OPT[0]:<br>0 = Disable digital input weak pull-down<br>1 = Enable digital input weak pull-down<br><br>When selecting pins for ADCIN or thermistor functions, the OPT[5:0] bits will be used as follows:<br>OPT[5:4]: If unused, set OPT[4] to 1.<br>OPT[3]: Whether the pin is used as ADC input:<br>0 = No<br>1 = Yes<br>OPT[2]: Whether the pin is used as Tsensor input with pull-up resistor:<br>0 = No<br>1 = Yes<br>OPT[1]:<br>0 = Not used for FET temperature protection<br>1 = Used for FET temperature protection<br>OPT[0]:<br>0 = Not used for cell temperature protection<br>1 = Used for cell temperature protection |
| spi_mosi_pin_fxn | [ 1: 0] | 0x0 | RW | These bits configure which function this pin is used for:<br>0 = SPI_MOSI or I2C address pin<br>1 = General Output<br>2 = Unused Not used |


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| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| | | | | 3 = ADC Input or Thermistor |

## 2.8 DCHG Pin Config Register

Address Offset: 0x07

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| dchg_opt | [ 7: 2] | 0x0 | RW | The options for pin functions selected by these bit configurations are as follows:<br>When 0x8e2a[0] = 0, DCHG acts as the shadow signal of CHG:<br>OPT[5]: --- Low-side CHG function, output is low when CHG is enabled; --- Low-side CHG function, output is high when CHG is enabled;<br>When 0x8e2a[0] = 1, DCHG acts as a regular GPIO or analog mapping:<br>Multi-function Pin Configuration:<br>OPT[5], OPT[4],OPT[3]<br>000: Default CHG function, default low.<br>100: Watchdog multiplexed output (reset the host).<br>101: General digital output (can be configured to use REG1 LDO or REG18 LDO for drive power supply, can be configured to weak pull-down to VSS or weak pull-up to REG1).<br>110: DCHG function (fault-related signal, can be used to control external protection circuits, used for interrupt control when the host manages CFETOFF to turn off CHGFET; configurable in NORMAL, SLEEP, DEEPSLEEP modes).<br>111: ADC or Tsensor input (unused for the 64-pin package).<br>Additional Configurations:<br>OPT[3]:<br>0 = Output high-level drive using REG1.<br>1 = Output high-level drive using REG18.<br>OPT[2]:<br>0 = Disable digital input (IE=0).<br>1 = Enable digital input (IE=1).<br>OPT[1]:<br>0 = Disable weak pull-up for digital input.<br>1 = Enable weak pull-up for digital input.<br>OPT[0]:<br>0 = Disable weak pull-down for digital input. |


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| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| | | | | 1 = Enable weak pull-down for digital input.<br><br>When selecting pins for ADCIN or thermistor functions, OPT[5:0] bits are used as follows:**<br>OPT[5:4]: If unused, set OPT[4] to 1.<br>OPT[3]: Is the pin used as ADC input<br>0 = No<br>1 = Yes<br>OPT[2]: Is the pin used as Tsensor input with pull-up resistor?<br>0 = No<br>1 = Yes<br>OPT[1]:<br>0 = Not used for FET temperature protection.<br>1 = Used for FET temperature protection.<br>OPT[0]:<br>0 = Not used for cell temperature protection.<br>1 = Used for cell temperature protection. |
| dchg_pin_fxn | [ 1: 0] | 0x0 | RW | These bits configure which function this pin is used for:<br>0 = Watchdog timer reset output.<br>1 = General Output.<br>2 = DCHG.<br>3 = ADC Input or Thermistor. |

## 2.9 DDSG Pin Config Register

Address Offset: 0x08

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| ddsg_opt | [ 7: 2] | 0x0 | RW | The options for pin functions selected by these bit configurations are as follows:<br>When 0x8e2a[1] = 0, DDSG acts as the shadow signal of DSG:<br>OPT[5]: --- Low-side DSG function, output is low when DSG is enabled; --- Low-side DSG function, output is high when DSG is enabled;<br>When 0x8e2a[1] = 1, DDSG acts as a regular GPIO or analog mapping:<br>Multi-function Pin Configuration:<br>OPT[5], OPT[4]:<br>000: Default DSG function, default low.<br>101: General digital output (can be configured to use REG1 |


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| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| | | | | LDO or REG18 LDO for drive power supply, can be configured to weak pull-down to VSS or weak pull-up to REG1).<br>110: DDSG function (fault-related signal, can be used to control external protection circuits, used for interrupt control when the host manages DFETOFF to turn off DSGFET; configurable in NORMAL, SLEEP, DEEPSLEEP modes).<br>111: ADC or Tsensor input (unused for the 64-pin package).<br>Additional Configurations:<br>OPT[3]:<br>0 = Output high-level drive using REG1.<br>1 = Output high-level drive using REG18.<br>OPT[2]:<br>0 = Disable digital input (IE=0).<br>1 = Enable digital input (IE=1).<br>OPT[1]:<br>0 = Disable weak pull-up for digital input.<br>1 = Enable weak pull-up for digital input.<br>OPT[0]:<br>0 = Disable weak pull-down for digital input.<br>1 = Enable weak pull-down for digital input.<br><br>When selecting pins for ADCIN or thermistor functions, OPT[5:0] bits are used as follows:<br>OPT[5:4]: If unused, set OPT[4] to 1.<br>OPT[3]: Is the pin used as ADC input?<br>0 = No<br>1 = Yes<br>OPT[2]: Is the pin used as Tsensor input with pull-up resistor?<br>0 = No<br>1 = Yes<br>OPT[1]:<br>0 = Not used for FET temperature protection.<br>1 = Used for FET temperature protection.<br>OPT[0]:<br>0 = Not used for cell temperature protection.<br>1 = Used for cell temperature protection. |
| ddsg_pin_fxn | [ 1: 0] | 0x0 | RW | ddsg_pin_fxn [1:0] 0x0 RW These bits configure which function this pin is used for:<br>0 = Reserved<br>1 = General Output |


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| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| | | | | 2 = DDSG<br>3 = ADC Input or Thermistor |

## 2.10 TS0_1 Config Register

Address Offset: 0x09

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| ts1_opt | [ 7: 4] | 0x0 | RW | OPT[3]:<br>Determines whether the pin is used as an ADC input:<br>0 = No<br>1 = Yes<br>OPT[2]:<br>Determines whether the pin is used as an external Tsensor input:<br>0 = No<br>1 = Yes<br>OPT[1]:<br>0 = Not used for FET temperature protection<br>1 = Used for FET temperature protection<br>OPT[0]:<br>0 = Not used for cell temperature protection<br>1 = Used for cell temperature protection<br>TS1 OPT[0] must write “1” the pin is used as an external Tsensor input. |
| ts0_opt | [ 3: 0] | 0x0 | RW | Reserved |

## 2.11 TS2_3 Config Register

Address Offset: 0x0A

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| ts3_opt | [ 7: 4] | 0x0 | RW | OPT[3]:<br>Determines whether the pin is used as an ADC input:<br>0 = No<br>1 = Yes<br>OPT[2]:<br>Determines whether the pin is used as an external Tsensor input:<br>0 = No<br>1 = Yes |


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| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| | | | | OPT[1]:<br>0 = Not used for FET temperature protection<br>1 = Used for FET temperature protection<br>OPT[0]:<br>0 = Not used for cell temperature protection<br>1 = Used for cell temperature protection |
| ts2_opt | [ 3: 0] | 0x0 | RW | OPT[3]:<br>Determines whether the pin is used as an ADC input:<br>0 = No<br>1 = Yes<br>OPT[2]:<br>Determines whether the pin is used as an external Tsensor input:<br>0 = No<br>1 = Yes<br>OPT[1]:<br>0 = Not used for FET temperature protection<br>1 = Used for FET temperature protection<br>OPT[0]:<br>0 = Not used for cell temperature protection<br>1 = Used for cell temperature protection |

## 2.12 PAD Filter En Register

Address Offset: 0x0E

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| da_ddsg_flt_en | [ 8: 8] | 0x0 | RW | 0: No filtering 1: Filtering enabled |
| da_dchg_flt_en | [ 7: 7] | 0x0 | RW | 0: No filtering 1: Filtering enabled |
| da_dfetoff_flt_en | [ 6: 6] | 0x0 | RW | 0: No filtering 1: Filtering enabled |
| da_cfetoff_flt_en | [ 5: 5] | 0x0 | RW | 0: No filtering 1: Filtering enabled |
| da_spics_flt_en | [ 4: 4] | 0x0 | RW | 0: No filtering 1: Filtering enabled |
| da_spimosi_flt_en | [ 3: 3] | 0x0 | RW | 0: No filtering 1: Filtering enabled |
| da_sda_flt_en | [ 2: 2] | 0x0 | RW | 0: No filtering 1: Filtering enabled |
| da_scl_flt_en | [ 1: 1] | 0x0 | RW | 0: No filtering 1: Filtering enabled |
| da_alert_flt_en | [ 0: 0] | 0x0 | RW | 0: No filtering 1: Filtering enabled |

## 2.13 PAD CFG Register

Address Offset: 0x10

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| pin_mode | [ 7: 7] | 0x1 | RW | Reserved |
| da_fuse_oe | [ 6: 6] | 0x0 | RW | 0 = FUSE pin configured as input pin<br>1 = FUSE pin configured as output pin |


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| | | | | |
|---|---|---|---|---|
| ts_res_sel | [ 5: 4] | 0x0 | RW | 0 = No pull-up<br>1 = Tsensor input, 18k pull-up<br>2 = Tsensor input, 180k pull-up<br>3 = Not used |
| busbar_sel | [ 3: 0] | 0x0 | RW | Reserved |


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# 3 Adc

Base Address: 0x8400

## 3.1 CELL1_GAIN Register

Address Offset: 0x00

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cell1_gain | [15:0] | 0x2ee0 | RW | Cell voltage calculations use a linear gain set by this value. Gain is factory-calibrated but can be modified by adjusting this value.<br>0 = Use factory-calibrated value.<br>All other values = Use specified value. |

## 3.2 CELL2_GAIN Register

Address Offset: 0x02

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cell2_gain | [15:0] | 0x2ee0 | RW | Cell voltage calculations use a linear gain set by this value. Gain is factory-calibrated but can be modified by adjusting this value.<br>0 = Use factory-calibrated value.<br>All other values = Use specified value. |

## 3.3 CELL3_GAIN Register

Address Offset: 0x04

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cell3_gain | [15:0] | 0x2ee0 | RW | Cell voltage calculations use a linear gain set by this value. Gain is factory-calibrated but can be modified by adjusting this value.<br>0 = Use factory-calibrated value.<br>All other values = Use specified value. |

## 3.4 CELL4_GAIN Register

Address Offset: 0x06

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cell4_gain | [15:0] | 0x2ee0 | RW | Cell voltage calculations use a linear gain set by this value. Gain is factory-calibrated but can be modified by adjusting this value.<br>0 = Use factory-calibrated value.<br>All other values = Use specified value. |

## 3.5 CELL5_GAIN Register

Address Offset: 0x08


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| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cell5_gain | [15:0] | 0x2ee0 | RW | Cell voltage calculations use a linear gain set by this value. Gain is factory-calibrated but can be modified by adjusting this value.<br>0 = Use factory-calibrated value.<br>All other values = Use specified value. |

## 3.6 CELL6_GAIN Register

Address Offset: 0x0A

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cell6_gain | [15:0] | 0x2ee0 | RW | Cell voltage calculations use a linear gain set by this value. Gain is factory-calibrated but can be modified by adjusting this value.<br>0 = Use factory-calibrated value.<br>All other values = Use specified value. |

## 3.7 CELL7_GAIN Register

Address Offset: 0x0C

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cell7_gain | [15:0] | 0x2ee0 | RW | Cell voltage calculations use a linear gain set by this value. Gain is factory-calibrated but can be modified by adjusting this value.<br>0 = Use factory-calibrated value.<br>All other values = Use specified value. |

## 3.8 CELL8_GAIN Register

Address Offset: 0x0E

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cell8_gain | [15:0] | 0x2ee0 | RW | Cell voltage calculations use a linear gain set by this value. Gain is factory-calibrated but can be modified by adjusting this value.<br>0 = Use factory-calibrated value.<br>All other values = Use specified value. |

## 3.9 CELL9_GAIN Register

Address Offset: 0x10

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cell9_gain | [15:0] | 0x2ee0 | RW | Cell voltage calculations use a linear gain set by this value. Gain is factory-calibrated but can be modified by adjusting this value.<br>0 = Use factory-calibrated value.<br>All other values = Use specified value |


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## 3.10 Reserve GAIN Register

Address Offset: 0x12

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| Reserve _gain | [15:0] | 0x2ee0 | RW | Cell voltage calculations use a linear gain set by this value. Gain is factory-calibrated but can be modified by adjusting this value.<br>0 = Use factory-calibrated value.<br>All other values = Use specified value |

## 3.11 CELL10_GAIN Register

Address Offset: 0x14

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cell10_gain | [15:0] | 0x2ee0 | RW | Cell voltage calculations use a linear gain set by this value. Gain is factory-calibrated but can be modified by adjusting this value.<br>0 = Use factory-calibrated value.<br>All other values = Use specified value |

## 3.12 CELL11_GAIN Register

Address Offset: 0x16

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cell11_gain | [15:0] | 0x2ee0 | RW | Cell voltage calculations use a linear gain set by this value. Gain is factory-calibrated but can be modified by adjusting this value.<br>0 = Use factory-calibrated value.<br>All other values = Use specified value |

## 3.13 CELL12_GAIN Register

Address Offset: 0x18

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cell12_gain | [15:0] | 0x2ee0 | RW | Cell voltage calculations use a linear gain set by this value. Gain is factory-calibrated but can be modified by adjusting this value.<br>0 = Use factory-calibrated value.<br>All other values = Use specified value |

## 3.14 CELL13_GAIN Register

Address Offset: 0x1A

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cell13_gain | [15:0] | 0x2ee0 | RW | Cell voltage calculations use a linear gain set by this value. Gain is factory-calibrated but can be modified by adjusting this value. |


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| | | | | |
|---|---|---|---|---|
| | | | | 0 = Use factory-calibrated value.<br>All other values = Use specified value |

## 3.15 CELL14_GAIN Register

Address Offset: 0x1C

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cell14_gain | [15:0] | 0x2ee0 | RW | Cell voltage calculations use a linear gain set by this value. Gain is factory-calibrated but can be modified by adjusting this value.<br>0 = Use factory-calibrated value.<br>All other values = Use specified value |

## 3.16 CELL15_GAIN Register

Address Offset: 0x1E

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cell15_gain | [15:0] | 0x2ee0 | RW | Cell voltage calculations use a linear gain set by this value. Gain is factory-calibrated but can be modified by adjusting this value.<br>0 = Use factory-calibrated value.<br>All other values = Use specified value |

## 3.17 CELL16_GAIN Register

Address Offset: 0x20

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cell16_gain | [15:0] | 0x2ee0 | RW | Cell voltage calculations use a linear gain set by this value. Gain is factory-calibrated but can be modified by adjusting this value.<br>0 = Use factory-calibrated value.<br>All other values = Use specified value |

## 3.18 PACK_GAIN Register

Address Offset: 0x22

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| pack_gain | [15:0] | 0x14c0 | RW | The PACK pin voltage calculation uses a linear gain set by this value. |

## 3.19 TOS_GAIN Register

Address Offset: 0x24

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| tos_gain | [15:0] | 0x14c0 | RW | The Top-of-Stack voltage calculation uses a linear gain set by this value. |


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## 3.20 LD_GAIN Register

Address Offset: 0x26

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| ld_gain | [15:0] | 0x14c0 | RW | The LD pin voltage calculation uses a linear gain set by this value. |

## 3.21 ADC_GAIN Register

Address Offset: 0x28

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| adc_gain | [15:0] | 0xfa0 | RW | Pins configured as ADCIN use this linear gain value to convert ADC counts to millivolts. |

## 3.22 CC_GAIN Register

Address Offset: 0x2C

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cc_gain | [19:0] | 0xbb8 | RW | This parameter sets the gain factor used to convert coulomb counter raw count measurements to current. The value is given by Calibration:Current:CC Gain = 30 / (Rsense in Ohm) / UserA.User calculates userA based on the actual connected resistance, referring to the porting guide. |

## 3.23 CC2_GAIN Register

Address Offset: 0x30

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cc2_gain | [11:0] | 0x0 | RW | This parameter sets the gain factor used to convert coulomb counter raw count measurements to passed charge. |

## 3.24 CC2_OFFSET Register

Address Offset: 0x32

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cc2_offset | [15:0] | 0x0 | RW | This parameter sets the offset factor used to convert coulomb counter raw count measurements to passed charge in normal mode.The unit is 0.05mA. |

## 3.25 VCELL_OFFSET Register

Address Offset: 0x34

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| vcell_offset | [15:0] | 0x0 | RW | This offset is subtracted from all cell voltage measurement values and can be used to calibrate out offset error. |

## 3.26 VDIV_OFFSET Register


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Address Offset: 0x36

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| vdiv_offset | [15:0] | 0x0 | RW | This offset is subtracted from all Stack Voltage(), PACK Pin Voltage(), and LD Pin Voltage() measurements and can be used to calibrate out offset error. |

## 3.27 ADC_OFFSET Register

Address Offset: 0x38

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| adc_offset | [15:0] | 0x0 | RW | This offset is subtracted from ADCIN measurements and can be used to calibrate out offset error. |

## 3.28 CC_OFFSET_SAMP Register

Address Offset: 0x3A

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cc_offset_samp | [15:0] | 0x40 | RW | the device provides an averaged count reading (cc2_avg_counts 0x260) over a configurable number of CC2 Counts samples. This parameter defines the number of samples that are accumulated before calculating a new average. |

## 3.29 BOARD_OFFSET Register

Address Offset: 0x3C

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| board_offset | [15:0] | 0x0 | RW | This board-level offset is subtracted from the coulomb counter conversion results in current calculations. To enable higher-resolution calibration, this value is divided by Calibration:Current Offset:Coulomb Counter Offset Samples before being used. This enables subtraction of fractional-count offsets. |

## 3.30 SLEEP_MODE_OFFSET Register

Address Offset: 0x3E

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| sleep_mode_offset | [7:0] | 0x0 | RW | The internal temperature calculation subs this offset to the result after gain is applied. |


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### 3.31 INT_GAIN Register

Address Offset: 0x40

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| int_gain | [15:0] | 0x2e66 | RW | The internal temperature calculation uses a linear gain set by this value. |

### 3.32 INTER_TEMP_OFFSET Register

Address Offset: 0x42

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| inter_temp_offset | [15:0] | 0xa5e | RW | This offset is subs to internal temperature measurements to increase accuracy. |

### 3.33 INT_MAX_AD Register

Address Offset: 0x44

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| int_max_ad | [15:0] | 0x1f40 | RW | The internal temperature calculation accuracy is limited when counts exceed a certain value. When the counts are greater than that value, a fixed value is reported for internal temperature. This parameter specifies that threshold. This value should not normally be changed. |

### 3.34 INT_MAX_TEMP Register

Address Offset: 0x46

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| int_max_temp | [15:0] | 0x7d0 | RW | When the internal temperature ADC measurement's counts exceed Calibration:Internal Temp Model:Int Maximum AD, this temperature is reported for internal temperature. This value should not normally be changed. |

### 3.35 COEFF_A1_18K Register

Address Offset: 0x48

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| coeff_a1_18k | [15:0] | 0x0 | RW | This is the coefficient of the 4th power element of the first temperature polynomial. |

### 3.36 COEFF_A2_18K Register

Address Offset: 0x4A

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| coeff_a2_18k | [15:0] | 0x0 | RW | This is the coefficient of the 3rd power element of the first temperature polynomial. |


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## 3.37 COEFF_A3_18K Register

Address Offset: 0x4C

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| coeff_a3_18k | [15:0] | 0x0 | RW | This is the coefficient of the 2nd power element of the first temperature polynomial. |

## 3.38 COEFF_A4_18K Register

Address Offset: 0x4E

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| coeff_a4_18k | [15:0] | 0x0 | RW | This is the coefficient of the 1st power element of the first temperature polynomial. |

## 3.39 COEFF_A1_180K Register

Address Offset: 0x50

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| coeff_a1_180k | [15:0] | 0x0 | RW | This is the coefficient of the 4th power element of the first temperature polynomial. |

## 3.40 COEFF_A2_180K Register

Address Offset: 0x52

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| coeff_a2_180k | [15:0] | 0x0 | RW | This is the coefficient of the 3rd power element of the first temperature polynomial. |

## 3.41 COEFF_A3_180K Register

Address Offset: 0x54

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| coeff_a3_180k | [15:0] | 0x0 | RW | This is the coefficient of the 2nd power element of the first temperature polynomial. |

## 3.42 COEFF_A4_180K Register

Address Offset: 0x56

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| coeff_a4_180k | [15:0] | 0x0 | RW | This is the coefficient of the 1st power element of the first temperature polynomial. |

## 3.43 COEFF_A1_EXT Register

Address Offset: 0x58

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| coeff_a1_ext | [15:0] | 0x0 | RW | This is the coefficient of the 4th power element of the first temperature polynomial. |


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## 3.44 COEFF_A2_EXT Register

Address Offset: 0x5A

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| coeff_a2_ext | [15:0] | 0x0 | RW | This is the coefficient of the 3rd power element of the first temperature polynomial. |

## 3.45 COEFF_A3_EXT Register

Address Offset: 0x5C

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| coeff_a3_ext | [15:0] | 0x0 | RW | This is the coefficient of the 2nd power element of the first temperature polynomial. |

## 3.46 COEFF_A4_EXT Register

Address Offset: 0x5E

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| coeff_a4_ext | [15:0] | 0x0 | RW | This is the coefficient of the 1st power element of the first temperature polynomial. |

## 3.47 CC_DEADBAND Register

Address Offset: 0x60

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cc_deadband | [7:0] | 0x9 | RW | To enable accurate charge accumulation, a deadband threshold is used to filter out signals below the expected noise floor. When the average coulomb counter output is below the deadband threshold, the charge is discarded instead of accumulated. This normally should not be changed. |

## 3.48 CC3_SAMP Register

Address Offset: 0x61

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cc3_samp | [7:0] | 0x50 | RW | The device provides an averaged current reading (CC3) over a configurable number of CC2 Current samples. This parameter defines the number of samples that are accumulated before calculating a new average. |

## 3.49 ADC_CFG Register

Address Offset: 0x62

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cadc_cic_skip | [7:6] | 0x0 | RW | curr adc cic skip num<br>0: cic skip 2 num |


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| | | | | |
|---|---|---|---|---|
| | | | | 1: cic skip 3 num<br>Others:reserve |
| vadc_cic_skip | [5:4] | 0x0 | RW | volt adc cic skip num<br>0: cic skip 2 num<br>1: cic skip 3 num<br>Others: reserve |
| cc1_normal_sel | [3:2] | 0x0 | RW | cc1 sampel sel for normal mode |
| cc1_sleep_sel | [1:0] | 0x0 | RW | cc1 sampel sel for sleep mode |

## 3.50 ADC_OSR_SEL Register

Address Offset: 0x63

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cadc_osr_normal_sel | [7:6] | 0x0 | RW | normal mode : curr adc osr sel |
| vadc_osr_normal_sel | [5:4] | 0x0 | RW | normal mode : volt adc osr sel |
| cadc_osr_sleep_sel | [3:2] | 0x0 | RW | sleep mode : curr adc osr sel |
| vadc_osr_sleep_sel | [1:0] | 0x0 | RW | sleep mode : volt adc osr sel |

## 3.51 ADC_CC2_RSENSE Register

Address Offset: 0x64

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cc2_rsense | [15:0] | 0x0 | RW | cc2 rsense |

## 3.52 CODE_GAIN Register

Address Offset: 0x66

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| code_gain | [15:0] | 0x4000 | RW | Thremistor adc code gain |

## 3.53 CODE_OFFSET Register

Address Offset: 0x68

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| code_offset | [15:0] | 0x0 | RW | Thremistor adc code offset |

## 3.54 ADC_EN Register

Address Offset: 0x6A

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| adc_rpt_ready | [7:7] | 0x0 | RO | 1: adc all cell volt/curr value ready, mcu can read. |
| adc_en | [0:0] | 0x1 | RW | ADC enable |


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## 3.55 CELL_GAIN_C Register

Address Offset: 0x6B

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cell_gain_c | [6:0] | 0x0 | RW | adc cell volt gain_c |

## 3.56 CELL_GAIN_B Register

Address Offset: 0x6C

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cell_gain_b | [10:0] | 0x0 | RW | adc cell volt gain_b |

## 3.57 CELL_GAIN_A Register

Address Offset: 0x6E

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cell_gain_a | [10:0] | 0x0 | RW | adc cell volt gain_a |

## 3.58 CC2_OFFSET_SLEEP Register

Address Offset: 0x70

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cc2_offset_sleep | [15:0] | 0x0 | RW | This parameter sets the offset factor used to convert coulomb counter raw count measurements to passed charge in sleep mode. |


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# 4 Protect

Base Address: 0x8800

## 4.1 SF_CTRL Register

Address Offset: 0x00

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| sf_en | [1:1] | 0x1 | RW | 0: primary protection disabled<br>1: primary protection enabled |
| da_pt_en | [0:0] | 0x1 | RW | 0: comparator for cov cuv occ ocd1 ocd2 disabled<br>1: comparator for cov cuv occ ocd1 ocd2 enabled |

## 4.2 LOAD_REMOVE_TH Register

Address Offset: 0x02

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| load_remove_th | [15:0] | 0x32 | RW | when load_detection is enabled if ld_pin_voltage is greater than this value , it means load removed. The uints is userV, default 8mV. |

## 4.3 FUSE_MIN_BLOW_FUSE_VOLTAGE Register

Address Offset: 0x04

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| min_blow_fuse_voltage | [15:0] | 0x1f4 | RW | The device will normally attempt to blow the fuse only if the stack voltage is above this threshold. If PACK_FUSE is set, pack voltage is used instead of stack voltage. However, if FET failure (CFETF or DFETF) is detected and FETF_FUSE is set, this voltage threshold is ignored. Range in 0~255. |

## 4.4 FUSE_BLOW_TIMEOUT Register

Address Offset: 0x06

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| fuse_blow_timeout | [7:0] | 0x1e | RW | When blowing the fuse, the device will assert the fuse blow output for this duration. Range in 0~255.<br>0: Disabled.<br>All other values : Drive fuse blow output for this many seconds when blowing the fuse. |


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## 4.5 DA_CONFIGURATION Register

Address Offset: 0x07

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| da_configuration | [7:0] | 0x5 | RW | [1:0] : reserved, must be configure b'01.<br>[2] : userV define To ensure the Top-of-Stack, PACK, and LD pin voltages fit in a signed 16-bit integer type, their units are configurable. This configurable unit is called user-volts and can be set to either centivolts or millivolts. For applications which will not exceed 32 Volts, millivolts may be used. Other applications should use centivolts to avoid saturating at 32767 mV.<br>0 = Millivolt (1 mV) units are selected for user-volts<br>1 = Centivolt (8 mV) units are selected for user-volts<br>[3] : TINT_EN enables the internal temperature source to be used as the "CellTemperature" for protections and logic that use minimum, maximum, or average temperature.<br>0:Internal temperature is not used for "Cell Temperature"<br>1:Internal temperature is used for "Cell Temperature"<br>[4] : TINT_FETT enables the internal temperature source to be used as the "FET Temperature". TINT_EN must also be set for this to apply. When TINT_EN = 0, this bit is ignored. When configured for FET temperature, it is not factored into minimum, maximum, and average cell temperature calculations.<br>0:Internal temperature is not used for "FET Temperature"<br>1:Internal temperature is used for "FET Temperature" if TINT_EN is also set Others : RESERVED. |

## 4.6 VCELL_MODE Register

Address Offset: 0x08

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| vcell_mode | [15:0] | 0x0 | RW | [15] : Cell 16 Mode<br>[14] : Cell 15 Mode<br>[13] : Cell 14 Mode<br>[12] : Cell 13 Mode<br>[11] : Cell 12 Mode<br>[10] : Cell 11 Mode<br>[9] : Cell 10 Mode<br>[8] : Reserve<br>[7] : Cell 9 Mode<br>[6] : Cell 8 Mode<br>[5] : Cell 7 Mode |


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| | | | | |
|---|---|---|---|---|
| | | | | [4] : Cell 6 Mode<br>[3] : Cell 5 Mode<br>[2] : Cell 4 Mode<br>[1] : Cell 3 Mode<br>[0] : Cell 2 Mode<br>each bit 1 : means this cell is in use<br>each bit 0 : means this cell is not in use Cell 1 is fixed in use |

**Note:** In cases where there are 10 or more strings, configure cell mode to leave the cell 10 empty. For example, the external pins short CELL16, CELL15, CELL14, and CELL13 together. Configure vcell_mode to 0001, 1110, 1111, 1111. For parts exceeding 10 cells, the pin cell names and register values differ by 1. That is, external vcell13 corresponds to internal register cell14.

## 4.7 PROTECTION_CONFIGURATION Register

Address Offset: 0x0A

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| scdl_curr_recov | [10:10] | 0x0 | RW | The SCD Latch fault may recover based on load removal detection on the LD pin, by host command, or by charge current. When the current-based recovery mechanism is not desired, it can be disabled by clearing this bit.<br>0: SCDL does not recover based on charge current.<br>1: SCDL recovers when current is greater than or equal to SCDL_Recovery_TH for SCDL_Recov_DELAY. |
| ocdl_curr_recov | [9:9] | 0x0 | RW | The OCD Latch fault may recover based on load removal detection on the LD pin, by host command, or by charge current. When the current-based recovery mechanism is not desired, it can be disabled by clearing this bit.<br>0: OCDL does not recover based on charge current.<br>1: OCDL recovers when current is greater than or equal to OCDL_Recovery_TH for OCDL_Recov_DELAY. |
| fetf_fuse | [8:8] | 0x0 | RW | When a Permanent Failure has been detected, the device may be configured to blow the fuse. This normally requires the voltage being above Min_Blow_Fuse_Voltage. When this bit is set and a FET failure occurs (CFETF or DFETF), the voltage requirement is bypassed.<br>0: Voltage must be above a threshold to blow the fuse when CFETF or DFETF trips.<br>1: If configured to blow the fuse and CFETF or DFETF occurs, fuse blow is attempted regardless of voltage. |


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| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| pack_fuse | [7:7] | 0x0 | RW | The fuse is typically located on the BAT side of the FETs, so the device monitors the Top of Stack voltage to determine if fuse blow is possible. However, some systems may place the fuse on the PACK side, in which case the PACK pin voltage should be monitored to determine if fuse blow is possible. Setting this bit causes the device to use the PACK pin voltage instead of the Top of Stack pin voltage for this comparison.<br>0: Top of Stack voltage must be above Min_Blow_Fuse_Voltage in order to blow the fuse.<br>1: PACK voltage must be above Min_Blow_Fuse_Voltage in order to blow the fuse. |
| pf_otp | [5:5] | 0x0 | RW | 0: Permanent Failure status is lost on reset.<br>1: Permanent Failure status is preserved in otp across reset when possible. |
| pf_fuse | [4:4] | 0x0 | RW | When a Permanent Failure occurs, the device may be configured to blow the fuse when voltage conditions are met.<br>0: Permanent Failure does not cause the device to blow the fuse.<br>1: Permanent Failure causes the device to blow the fuse. |
| pf_dpslp | [3:3] | 0x0 | RW | 0:Device does not automatically enter DEEPSLEEP mode when a Permanent Failure occurs.<br>1: Device automatically enters DEEPSLEEP mode when a Permanent Failure occurs. |
| pf_regs | [2:2] | 0x0 | RW | 0: Permanent Failure does not cause the device to turn the regulators off.<br>1: Permanent Failure causes the device to turn the regulators off. |
| pf_fets | [1:1] | 0x1 | RW | 0: Permanent Failure does not cause the device to turn the FETs off.<br>1: Permanent Failure causes the device to turn the FETs off. |

## 4.8 ENABLED_PROTECTIONS_A Register

Address Offset: 0x0C

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| scd_prot_en | [7:7] | 0x1 | RW | Short Circuit in Discharge Protection<br>0: Disabled<br>1: Enabled |
| ocd2_prot_en | [6:6] | 0x0 | RW | Overcurrent in Discharge 2nd Tier Protection<br>0: Disabled<br>1: Enabled |


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| | | | | |
|---|---|---|---|---|
| ocd1_prot_en | [5:5] | 0x0 | RW | Overcurrent in Discharge 1st Tier Protection<br>0: Disabled<br>1: Enabled |
| occ_prot_en | [4:4] | 0x0 | RW | Overcurrent in Charge Protection<br>0: Disabled<br>1: Enabled |
| cov_prot_en | [3:3] | 0x1 | RW | Cell Overvoltage Protection<br>0: Disabled<br>1: Enabled |
| cuv_prot_en | [2:2] | 0x0 | RW | Cell Undervoltage Protection<br>0: Disabled<br>1: Enabled |

## 4.9 ENABLED_PROTECTIONS_B Register

Address Offset: 0x0D

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| otf_prot_en | [7:7] | 0x0 | RW | FET Overtemperature<br>0: Disabled<br>1: Enabled |
| otint_prot_en | [6:6] | 0x0 | RW | Internal Overtemperature<br>0: Disabled<br>1: Enabled |
| otd_prot_en | [5:5] | 0x0 | RW | Overtemperature in Discharge<br>0: Disabled<br>1: Enabled |
| otc_prot_en | [4:4] | 0x0 | RW | Overtemperature in Charge<br>0: Disabled<br>1: Enabled |
| utint_prot_en | [2:2] | 0x0 | RW | Internal Undertemperature<br>0: Disabled<br>1: Enabled |
| utd_prot_en | [1:1] | 0x0 | RW | Undertemperature in Discharge<br>0: Disabled<br>1: Enabled |
| utc_prot_en | [0:0] | 0x0 | RW | Undertemperature in Charge<br>0: Disabled<br>1: Enabled |


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## 4.10 ENABLED_PROTECTIONS_C Register

Address Offset: 0x0E

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| ocd3_prot_en | [7:7] | 0x0 | RW | Overcurrent in Discharge 3rd Tier Protection<br>0: Disabled<br>1: Enabled |
| scdl_prot_en | [6:6] | 0x0 | RW | Short Circuit in Discharge Latch<br>0: Disabled<br>1: Enabled |
| ocdl_prot_en | [5:5] | 0x0 | RW | Overcurrent in Discharge Latch<br>0: Disabled<br>1: Enabled |
| covl_prot_en | [4:4] | 0x0 | RW | Cell Overvoltage Latch<br>0: Disabled<br>1: Enabled |
| pto_prot_en | [2:2] | 0x0 | RW | Precharge Timeout<br>0: Disabled<br>1: Enabled |
| hwdf_prot_en | [1:1] | 0x0 | RW | Host Watchdog Fault<br>0: Disabled<br>1: Enabled |

## 4.11 CHG_FET_PROTECTIONS_A Register

Address Offset: 0x0F

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| chg_fet_scd_en | [7:7] | 0x1 | RW | Short Circuit in Discharge Protection<br>0: CHG FET is not disabled when protection is triggered.<br>1: CHG FET is disabled when protection is triggered. |
| chg_fet_occ_en | [4:4] | 0x1 | RW | Overcurrent in Charge Protection<br>0: CHG FET is not disabled when protection is triggered.<br>1: CHG FET is disabled when protection is triggered. |
| chg_fet_cov_en | [3:3] | 0x1 | RW | Cell Overvoltage Protection<br>0: CHG FET is not disabled when protection is triggered.<br>1: CHG FET is disabled when protection is triggered. |

## 4.12 CHG_FET_PROTECTIONS_B Register

Address Offset: 0x10

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| chg_fet_otf_en | [7:7] | 0x1 | RW | FET Overtemperature<br>0: CHG FET is not disabled when protection is triggered.<br>1: CHG FET is disabled when protection is triggered. |


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| | | | | |
|---|---|---|---|---|
| chg_fet_otint_en | [6:6] | 0x1 | RW | Internal Overtemperature<br>0: CHG FET is not disabled when protection is triggered.<br>1: CHG FET is disabled when protection is triggered. |
| chg_fet_otc_en | [4:4] | 0x1 | RW | Overtemperature in Charge<br>0: CHG FET is not disabled when protection is triggered.<br>1: CHG FET is disabled when protection is triggered. |
| chg_fet_utint_en | [2:2] | 0x1 | RW | Internal Undertemperature<br>0: CHG FET is not disabled when protection is triggered.<br>1: CHG FET is disabled when protection is triggered. |
| chg_fet_utc_en | [0:0] | 0x1 | RW | Undertemperature in Charge<br>0: CHG FET is not disabled when protection is triggered.<br>1: CHG FET is disabled when protection is triggered. |

## 4.13 CHG_FET_PROTECTIONS_C Register

Address Offset: 0x11

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| chg_fet_scdl_en | [6:6] | 0x1 | RW | Short Circuit in Discharge Latch<br>0: CHG FET is not disabled when protection is triggered.<br>1: CHG FET is disabled when protection is triggered. |
| chg_fet_covl_en | [4:4] | 0x1 | RW | Cell Overvoltage Latch<br>0: CHG FET is not disabled when protection is triggered.<br>1: CHG FET is disabled when protection is triggered. |
| chg_fet_pto_en | [2:2] | 0x1 | RW | Precharge Timeout<br>0: CHG FET is not disabled when protection is triggered.<br>1: CHG FET is disabled when protection is triggered. |
| chg_fet_hwdf_en | [1:1] | 0x1 | RW | Host Watchdog Fault<br>0: CHG FET is not disabled when protection is triggered.<br>1: CHG FET is disabled when protection is triggered. |

## 4.14 DSG_FET_PROTECTIONS_A Register

Address Offset: 0x12

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| dsg_fet_scd_en | [7:7] | 0x1 | RW | Short Circuit in Discharge Protection<br>0: DSG FET is not disabled when protection is triggered.<br>1: DSG FET is disabled when protection is triggered. |
| dsg_fet_ocd2_en | [6:6] | 0x1 | RW | Overcurrent in Discharge 2nd Tier Protection<br>0: DSG FET is not disabled when protection is triggered.<br>1: DSG FET is disabled when protection is triggered. |
| dsg_fet_ocd1_en | [5:5] | 0x1 | RW | Overcurrent in Discharge 1st Tier Protection<br>0: DSG FET is not disabled when protection is triggered.<br>1: DSG FET is disabled when protection is triggered. |


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| | | | | |
|---|---|---|---|---|
| dsg_fet_cuv_en | [2:2] | 0x1 | RW | Cell Undervoltage Protection<br>0: DSG FET is not disabled when protection is triggered.<br>1: DSG FET is disabled when protection is triggered. |

## 4.15 DSG_FET_PROTECTIONS_B Register

Address Offset: 0x13

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| dsg_fet_otf_en | [7:7] | 0x1 | RW | FET Overtemperature<br>0: DSG FET is not disabled when protection is triggered.<br>1: DSG FET is disabled when protection is triggered. |
| dsg_fet_otint_en | [6:6] | 0x1 | RW | Internal Overtemperature<br>0: DSG FET is not disabled when protection is triggered.<br>1: DSG FET is disabled when protection is triggered. |
| dsg_fet_otd_en | [5:5] | 0x1 | RW | Overtemperature in Discharge<br>0: DSG FET is not disabled when protection is triggered.<br>1: DSG FET is disabled when protection is triggered. |
| dsg_fet_utint_en | [2:2] | 0x1 | RW | Internal Undertemperature<br>0: DSG FET is not disabled when protection is triggered.<br>1: DSG FET is disabled when protection is triggered. |
| dsg_fet_utd_en | [1:1] | 0x1 | RW | Undertemperature in Discharge<br>0: DSG FET is not disabled when protection is triggered.<br>1: DSG FET is disabled when protection is triggered. |

## 4.16 DSG_FET_PROTECTIONS_C Register

Address Offset: 0x14

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| dsg_fet_ocd3_en | [7:7] | 0x1 | RW | Overcurrent in Discharge 3rd Tier Protection<br>0: DSG FET is not disabled when protection is triggered.<br>1: DSG FET is disabled when protection is triggered. |
| dsg_fet_scdl_en | [6:6] | 0x1 | RW | Short Circuit in Discharge Latch<br>0: DSG FET is not disabled when protection is triggered.<br>1: DSG FET is disabled when protection is triggered. |
| dsg_fet_ocdl_en | [5:5] | 0x1 | RW | Overcurrent in Discharge Latch<br>0: DSG FET is not disabled when protection is triggered.<br>1: DSG FET is disabled when protection is triggered. |
| dsg_fet_hwdf_en | [1:1] | 0x1 | RW | Host Watchdog Fault<br>0: DSG FET is not disabled when protection is triggered.<br>1: DSG FET is disabled when protection is triggered. |

## 4.17 SF_ALERT_MASK_A Register

Address Offset: 0x15

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|


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| | | | | |
|---|---|---|---|---|
| scd_sf_alert_mask | [7:7] | 0x1 | RW | When the bitwise of this bit and SCD SafetyAlert is nonzero AlarmRawStatus()[MSK_SFALERT] is set. |
| ocd2_sf_alert_mask | [6:6] | 0x1 | RW | When the bitwise of this bit and OCD2 SafetyAlert is nonzero AlarmRawStatus()[MSK_SFALERT] is set. |
| ocd1_sf_alert_mask | [5:5] | 0x1 | RW | When the bitwise of this bit and OCD1 SafetyAlert is nonzero AlarmRawStatus()[MSK_SFALERT] is set. |
| occ_sf_alert_mask | [4:4] | 0x1 | RW | When the bitwise of this bit and OCC SafetyAlert is nonzero AlarmRawStatus()[MSK_SFALERT] is set. |
| cov_sf_alert_mask | [3:3] | 0x1 | RW | When the bitwise of this bit and COV SafetyAlert is nonzero AlarmRawStatus()[MSK_SFALERT] is set. |
| cuv_sf_alert_mask | [2:2] | 0x1 | RW | When the bitwise of this bit and CUV SafetyAlert is nonzero AlarmRawStatus()[MSK_SFALERT] is set. |

## 4.18 SF_ALERT_MASK_B Register

Address Offset: 0x16

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| otf_sf_alert_mask | [7:7] | 0x1 | RW | When the bitwise of this bit and OTF SafetyAlert is nonzero AlarmRawStatus()[MSK_SFALERT] is set. |
| otint_sf_alert_mask | [6:6] | 0x1 | RW | When the bitwise of this bit and OTINT SafetyAlert is nonzero AlarmRawStatus()[MSK_SFALERT] is set. |
| otd_sf_alert_mask | [5:5] | 0x1 | RW | When the bitwise of this bit and OTD SafetyAlert is nonzero AlarmRawStatus()[MSK_SFALERT] is set. |
| otc_sf_alert_mask | [4:4] | 0x1 | RW | When the bitwise of this bit and OTC SafetyAlert is nonzero AlarmRawStatus()[MSK_SFALERT] is set. |
| utint_sf_alert_mask | [2:2] | 0x1 | RW | When the bitwise of this bit and UTINT SafetyAlert is nonzero AlarmRawStatus()[MSK_SFALERT] is set. |
| utd_sf_alert_mask | [1:1] | 0x1 | RW | When the bitwise of this bit and UTD SafetyAlert is nonzero AlarmRawStatus()[MSK_SFALERT] is set. |
| utc_sf_alert_mask | [0:0] | 0x1 | RW | When the bitwise of this bit and UTC SafetyAlert is nonzero AlarmRawStatus()[MSK_SFALERT] is set. |

## 4.19 SF_ALERT_MASK_C Register

Address Offset: 0x17

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| ocd3_sf_alert_mask | [7:7] | 0x1 | RW | When the bitwise of this bit and OCD3 SafetyAlert is nonzero AlarmRawStatus()[MSK_SFALERT] is set. |
| scdl_sf_alert_mask | [6:6] | 0x1 | RW | When the bitwise of this bit and SCDL SafetyAlert is nonzero AlarmRawStatus()[MSK_SFALERT] is set. |
| ocdl_sf_alert_mask | [5:5] | 0x1 | RW | When the bitwise of this bit and OCDL SafetyAlert is nonzero AlarmRawStatus()[MSK_SFALERT] is set. |
| covl_sf_alert_mask | [4:4] | 0x1 | RW | When the bitwise of this bit and COVL SafetyAlert is nonzero AlarmRawStatus()[MSK_SFALERT] is set. |


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| | | | | |
|---|---|---|---|---|
| pto_sf_alert_mask | [2:2] | 0x1 | RW | When the bitwise of this bit and PTO SafetyAlert is nonzero AlarmRawStatus()[MSK_SFALERT] is set. |

## 4.20 PF_ALERT_MASK_A Register

Address Offset: 0x18

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cudep_pf_alert_mask | [7:7] | 0x0 | RW | When the bitwise of this bit and CUDEP SafetyAlert is nonzero AlarmRawStatus()[MSK_PFALERT] is set. |
| sotf_pf_alert_mask | [6:6] | 0x1 | RW | When the bitwise of this bit and SOTF SafetyAlert is nonzero AlarmRawStatus()[MSK_PFALERT] is set. |
| sot_pf_alert_mask | [4:4] | 0x1 | RW | When the bitwise of this bit and SOT SafetyAlert is nonzero AlarmRawStatus()[MSK_PFALERT] is set. |
| socd_pf_alert_mask | [3:3] | 0x1 | RW | When the bitwise of this bit and SOCD SafetyAlert is nonzero AlarmRawStatus()[MSK_PFALERT] is set. |
| socc_pf_alert_mask | [2:2] | 0x1 | RW | When the bitwise of this bit and SOCC SafetyAlert is nonzero AlarmRawStatus()[MSK_PFALERT] is set. |
| sov_pf_alert_mask | [1:1] | 0x1 | RW | When the bitwise of this bit and SOV SafetyAlert is nonzero AlarmRawStatus()[MSK_PFALERT] is set. |
| suv_pf_alert_mask | [0:0] | 0x1 | RW | When the bitwise of this bit and SUV SafetyAlert is nonzero AlarmRawStatus()[MSK_PFALERT] is set. |

## 4.21 PF_ALERT_MASK_B Register

Address Offset: 0x19

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| scdl_pf_alert_mask | [7:7] | 0x1 | RW | When the bitwise of this bit and SCDL SafetyAlert is nonzero AlarmRawStatus()[MSK_PFALERT] is set. |
| vima_pf_alert_mask | [4:4] | 0x1 | RW | When the bitwise of this bit and VIMA SafetyAlert is nonzero AlarmRawStatus()[MSK_PFALERT] is set. |
| vimr_pf_alert_mask | [3:3] | 0x1 | RW | When the bitwise of this bit and VIMR SafetyAlert is nonzero AlarmRawStatus()[MSK_PFALERT] is set. |
| dlvl_pf_alert_mask | [2:2] | 0x1 | RW | When the bitwise of this bit and DLVL SafetyAlert is nonzero AlarmRawStatus()[MSK_PFALERT] is set. |
| dfetf_pf_alert_mask | [1:1] | 0x1 | RW | When the bitwise of this bit and DFETF SafetyAlert is nonzero AlarmRawStatus()[MSK_PFALERT] is set. |
| cfetf_pf_alert_mask | [0:0] | 0x1 | RW | When the bitwise of this bit and CFETF SafetyAlert is nonzero AlarmRawStatus()[MSK_PFALERT] is set. |


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## 4.22 PF_ALERT_MASK_C Register

Address Offset: 0x1A

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| hwmx_pf_alert_mask | [6:6] | 0x0 | RW | When the bitwise of this bit and HWMX SafetyAlert is nonzero AlarmRawStatus()[MSK_PFALERT] is set. |
| vssf_pf_alert_mask | [5:5] | 0x0 | RW | When the bitwise of this bit and VSSF SafetyAlert is nonzero AlarmRawStatus()[MSK_PFALERT] is set. |
| vref_pf_alert_mask | [4:4] | 0x0 | RW | When the bitwise of this bit and VREF SafetyAlert is nonzero AlarmRawStatus()[MSK_PFALERT] is set. |
| lfof_pf_alert_mask | [3:3] | 0x0 | RW | When the bitwise of this bit and LFOF SafetyAlert is nonzero AlarmRawStatus()[MSK_PFALERT] is set. |

## 4.23 PF_ALERT_MASK_D Register

Address Offset: 0x1B

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| tosf_pf_alert_mask | [0:0] | 0x0 | RW | When the bitwise of this bit and TOSF SafetyAlert is nonzero AlarmRawStatus()[MSK_PFALERT] is set. |

## 4.24 BODY_DIODE_THRESHOLD Register

Address Offset: 0x1C

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| body_diode_th | [15:0] | 0x32 | RW | To minimize power dissipation in the FET body diode, the FET is turned on when reverse current is detected and the other FET is on. When measured discharge current is greater in magnitude than Settings:Protection:Body Diode Threshold and the DSG FET or PDSG FET is on, the CHG FET is turned on, and the PCHG FET is turned off. When measured charge current is greater than BODY_DIODE_TH and the CHG FET or PCHG FET is on, the DSG FET is turned on, and the PDSG FET is turned off. When in parallel FET mode (SFET = 0), body diode protection is disabled and a FET will not be turned on in response to reverse current. |


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## 4.25 DEFAULT_ALARM_MASK Register

Address Offset: 0x1E

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| alm_en | [15:0] | 0xf800 | RWHW | [15] SSBC Setting this bit causes the Alarm Status()[SSBC] to be set and latched when Alarm Raw Status()[SSBC] is asserted.<br>[14] SSA Setting this bit causes the Alarm Status()[SSA] to be set and latched when Alarm Raw Status()[SSA] is asserted.<br>[13] PF Setting this bit causes the Alarm Status()[PF] to be set and latched when Alarm Raw Status()[PF] is asserted.<br>[12] MSK_SFALERT Setting this bit causes the Alarm Status()[MSK_SFALERT] to be set and latched when Alarm Raw Status()[MSK_SFALERT] is asserted.<br>[11] MSK_PFALERT Setting this bit causes the Alarm Status()[MSK_PFALERT] to be set and latched when Alarm Raw Status()[MSK_PFALERT] is asserted.<br>[10] INITSTART Setting this bit causes the Alarm Status()[INITSTART] to be set and latched when Alarm Raw Status()[INITSTART] is asserted.<br>[9] INITCOMP Setting this bit causes the Alarm Status()[INITCOMP] to be set and latched when Alarm Raw Status()[INITCOMP] is asserted.<br>[8] MSK_SFFAULT Setting this bit causes the Alarm Status()[MSK_SFALERT] to be set and latched when Alarm Raw Status()[MSK_SFALERT] is asserted.<br>[7] FULLSCAN Setting this bit causes the Alarm Status()[FULLSCAN] to be set and latched when Alarm Raw Status()[FULLSCAN] is asserted.<br>[6] XCHG Setting this bit causes the Alarm Status()[XCHG] to be set and latched when Alarm Raw Status()[XCHG] is asserted.<br>[5] XDSG Setting this bit causes the Alarm Status()[XDSG] to be set and latched when Alarm Raw Status()[XDSG] is asserted.<br>[4] SHUTV Setting this bit causes the Alarm Status()[SHUTV] to be set and latched when Alarm Raw Status()[SHUTV] is asserted.<br>[3] FUSE Setting this bit causes the Alarm Status()[FUSE] to be set and latched when Alarm Raw Status()[FUSE] is asserted.<br>[2] CB Setting this bit causes the Alarm Status()[CB] to be set and latched when Alarm Raw Status()[CB] is asserted. |


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| | | | | |
|---|---|---|---|---|
| | | | | [1] ADSCAN Setting this bit causes the Alarm Status()[ADSCAN] to be set and latched when Alarm Raw Status()[ADSCAN] is asserted.<br>[0] WAKE Setting this bit causes the Alarm Status()[WAKE] to be set and latched when Alarm Raw Status()[WAKE] is asserted. |

## 4.26 ENABLED_PF_A Register

Address Offset: 0x20

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cudep_pf_en | [7:7] | 0x0 | RW | Copper Deposition Permanent Fail<br>0: Disabled<br>1: Enabled |
| sotf_pf_en | [6:6] | 0x0 | RW | Safety Overtemperature FET Permanent Fail<br>0: Disabled<br>1: Enabled |
| sot_pf_en | [4:4] | 0x0 | RW | Safety Overtemperature Permanent Fail<br>0: Disabled<br>1: Enabled |
| socd_pf_en | [3:3] | 0x0 | RW | Safety Overcurrent in Discharge Permanent Fail<br>0: Disabled<br>1: Enabled |
| socc_pf_en | [2:2] | 0x0 | RW | Safety Overcurrent in Charge Permanent Fail<br>0: Disabled<br>1: Enabled |
| sov_pf_en | [1:1] | 0x0 | RW | Safety Cell Overvoltage Permanent Fail<br>0: Disabled<br>1: Enabled |
| suv_pf_en | [0:0] | 0x0 | RW | Safety Cell Undervoltage Permanent Fail<br>0: Disabled<br>1: Enabled |

## 4.27 ENABLED_PF_B Register

Address Offset: 0x21

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| scdl_pf_en | [7:7] | 0x0 | RW | Short Circuit in Discharge Latch Permanent Fail<br>0: Disabled<br>1: Enabled |
| vima_pf_en | [4:4] | 0x0 | RW | Voltage Imbalance Active Permanent Fail<br>0: Disabled<br>1: Enabled |
| vimr_pf_en | [3:3] | 0x0 | RW | Voltage Imbalance at Rest Permanent Fail<br>0: Disabled |


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| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| | | | | 1: Enabled |
| dlvl_pf_en | [2:2] | 0x0 | RW | Second Level Protector Permanent Fail<br>0: Disabled<br>1: Enabled |
| dfetf_pf_en | [1:1] | 0x0 | RW | Discharge FET Permanent Fail<br>0: Disabled<br>1: Enabled |
| cfetf_pf_en | [0:0] | 0x0 | RW | Charge FET Permanent Fail<br>0: Disabled<br>1: Enabled |

## 4.28 ENABLED_PF_C Register

Address Offset: 0x22

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cmdf_pf_en | [7:7] | 0x0 | RW | Commanded Permanent Fail<br>0: Disabled<br>1: Enabled |
| hwmx_pf_en | [6:6] | 0x0 | RW | Internal Stuck Hardware Mux Permanent Fail<br>0: Disabled<br>1: Enabled |
| vssf_pf_en | [5:5] | 0x0 | RW | Internal VSS Measurement Permanent Fail<br>0: Disabled<br>1: Enabled |
| vref_pf_en | [4:4] | 0x0 | RW | Internal Voltage Reference Permanent Fail<br>0: Disabled<br>1: Enabled |
| lfof_pf_en | [3:3] | 0x0 | RW | Internal LFO Permanent Fail<br>0: Disabled<br>1: Enabled |
| irmf_pf_en | [2:2] | 0x1 | RW | No used |
| drmf_pf_en | [1:1] | 0x1 | RW | No used |
| otpf_pf_en | [0:0] | 0x1 | RW | OTP Memory Permanent Fail<br>0: Disabled<br>1: Enabled |

## 4.29 ENABLED_PF_D Register

Address Offset: 0x23

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| tosf_pf_en | [0:0] | 0x0 | RW | Top of Stack vs Cell Sum Permanent Fail<br>0: Disabled<br>1: Enabled |


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## 4.30 FET_OPTIONS Register

Address Offset: 0x24

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| fet_init_off | [5:5] | 0x0 | RW | 0: Default host FET control state allows FETs to be on<br>1: Default host FET control state forces FETs off |
| pdsg_en | [4:4] | 0x0 | RW | 0: PDSG FET is not used<br>1: PDSG FET is turned on before DSG |
| fet_ctrl_en | [3:3] | 0x1 | RW | 0: FETs will not be turned on<br>1: FETs are controlled by the device |
| host_fet_en | [2:2] | 0x1 | RW | 0: Host FET control commands are ignored<br>1: Host FET control commands are allowed |
| sleepchg | [1:1] | 0x0 | RW | 0: CHG FET is turned off in SLEEP mode<br>1: CHG FET may be enabled in SLEEP mode |
| sfet | [0:0] | 0x1 | RW | 0: Parallel FET mode: Body diode protection is disabled<br>1: Series FET mode: Body diode protection is enabled |

## 4.31 CHG_PUMP_CONTROL Register

Address Offset: 0x25

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| sleep_cpen | [3:3] | 0x1 | RW | 0: Charge pumps for FET drivers are disabled in sleep mode<br>1: Charge pumps for FET drivers are enabled in sleep mode |
| sfmode_sleep | [2:2] | 0x0 | RW | 0: Source-follower mode is not enabled on the DSG FET driver<br>1: Source-follower mode is enabled on the DSG FET driver while in SLEEP mode |
| lven | [1:1] | 0x0 | RW | This bit selects the charge pump overdrive level.<br>0: Charge pump low overdrive level (5.5 V) is selected<br>1: Charge pump high overdrive level (11 V) is selected |
| normal_cpen | [0:0] | 0x1 | RW | 0: Charge pumps for FET drivers are disabled in normal mode<br>1: Charge pumps for FET drivers are enabled in normal mode |

## 4.32 PRECHARGE_START_VOLTAGE Register

Address Offset: 0x26

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| pchg_start_voltage | [15:0] | 0x0 | RW | PRECHARGE mode can be used to provide lower-current charging through the PCHG FET instead of the CHG FET for an under-voltage battery. When the minimum cell voltage is less than this threshold, PRECHARGE mode is activated. To disable PRECHARGE mode, set this value to 0. |


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## 4.33 PRECHARGE_STOP_VOLTAGE Register

Address Offset: 0x28

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| pchg_stop_voltage | [15:0] | 0x0 | RW | PRECHARGE mode can be used to provide lower-current charging through the PCHG FET instead of the CHG FET for an under-voltage battery. When the minimum cell voltage is greater than or equal to this threshold, PRECHARGE mode is deactivated. To disable PRECHARGE mode, set this value to 0. |

## 4.34 PREDISCHARGE_TIMEOUT Register

Address Offset: 0x2A

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| pdsg_timeout | [7:0] | 0x5 | RW | When PREDISCHARGE mode is enabled, the maxiumum duration of PREDISCHARGE mode can be set in 10-ms increments.<br>0: No timeout. Predischarge mode is exited when voltage conditions are met.<br>All other values: Predischarge mode is exited and DSG is turned on after configured timeout. |

## 4.35 PREDISCHARGE_STOP_DELTA Register

Address Offset: 0x2B

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| pdsg_stop_delta_en | [7:7] | 0 | RW | Bit[7]:<br>0: Predischarge voltage check disabled. Predischarge mode is exited for timeout.<br>All other values:Predischarge mode exited when the LD pin voltage is measured within this delta of the top-of-stack voltage.<br>Bit[6]:<br>0: Predischarge time out check disabled.<br>1: Predischarge mode is exited when reach preset pdsg timeout.<br>[7:6]=00, cannot exit pre-charge;<br>[7:6]=11, exit pre-charge as long as one of the conditions is met;<br>[7:6]=01, exit pre-charge when Time out condition is met;<br>[7:6]=10, exit pre-charge when LD voltage condition is met. |
| pdsg timeout en | [6:6] | 0 | RW | |
| pdsg stop delta | [5:0] | 0x32 | RW | 128mV/LSB. Predischarge mode exited when the LD pin voltage is measured within this delta of the top-of-stack voltage. |


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## 4.36 CUV_THRESHOLD Register

Address Offset: 0x2C

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cuv_th | [8:0] | 0x78 | RW | This parameter sets the Cell Undervoltage Protection threshold in units of 20mV with 1 offset. For example, setting the decimal value to 99 corresponds to a CUV voltage of 2000mV. |

## 4.37 CUV_DELAY Register

Address Offset: 0x2E

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cuv_delay | [15:0] | 0x4a | RW | This parameter sets the Cell Undervoltage Protection delay in units of comparator loops(each loop is 3.9ms) with a 2 loops offset. |

## 4.38 CUV_RECOVERY_HYSTERESIS Register

Address Offset: 0x30

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cuv_recov_hys | [7:0] | 0x2 | RW | This parameter sets the Cell Undervoltage Protection hysteresis threshold in units of 20mV. The minimum cell voltage must be greater than or equal to the CUV threshold plus this hysteresis for RECOV_DELAY to recover from a CUV condition. |

## 4.39 COV_THRESHOLD Register

Address Offset: 0x32

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cov_th | [8:0] | 0xdc | RW | This parameter sets the Cell Overvoltage Protection threshold in units of 20mV with 1 offset. |

## 4.40 COV_DELAY Register

Address Offset: 0x34

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cov_delay | [15:0] | 0x4a | RW | This parameter sets the Cell Overvoltage Protection delay in units of comparator loops(each loop is 3.9ms) with a 2 loops offset. |

## 4.41 COV_RECOVERY_HYSTERESIS Register

Address Offset: 0x36

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|


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| | | | | |
|---|---|---|---|---|
| cov_recov_hys | [7:0] | 0x2 | RW | This parameter sets the Cell Overvoltage Protection hysteresis threshold in units of 20mV. The maximum cell voltage must be less than or equal to the COV threshold minus this hysteresis for RECOV_DELAY to recover from a COV condition. |

## 4.42 COVL_LATCH_LIMIT Register

Address Offset: 0x37

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| covl_latch_limit | [7:0] | 0x2 | RW | Whenever a COV event occurs, the COV latch counter is incremented. If the latch counter reaches this limit, the Cell Overvoltage Latch Protection is triggered. This protection allows for a longer recovery time when repeated COV events occur because the action of turning the CHG FET off cause the voltage to drop enough to recover. The value of the parameter covl_dec_delay must be less than covl_recov_delay. |

## 4.43 COVL_COUNTER_DEC_DELAY Register

Address Offset: 0x38

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| covl_dec_delay | [7:0] | 0xa | RW | Whenever a COV event occurs, the COV latch counter is incremented. While the counter is nonzero, it will be decremented after the delay set by this parameter. The step is with unit Second. |

## 4.44 COVL_RECOVERY_TIME Register

Address Offset: 0x39

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| covl_recov_delay | [7:0] | 0xf | RW | This parameter sets the delay after which the Cell Overvoltage Latch Protection will recover. The step is with unit Second. |

## 4.45 OCC_THRESHOLD Register

Address Offset: 0x3A

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| occ_th | [8:0] | 0x5 | RW | This parameter sets the Overcurrent in Charge Protection threshold for the sense resistor voltage in units of 2mV with 3mV offset . |

## 4.46 OCC_DELAY Register

Address Offset: 0x3C


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| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| occ_delay | [7:0] | 0x4 | RW | This parameter sets the Cell Overcurrent in Charge Protection delay in units of comparator loops(each loop is 3.9ms) with a 2 loops offset. |

## 4.47 OCC_RECOVERY_THRESHOLD Register

Address Offset: 0x40

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| occ_recov_th | [15:0] | 0xff38 | RW | This sets the recovery threshold for the Overcurrent in Charge Protection. If measured current is less than or equal to this threshold for RECOVER_DELAY, recovery occurs. The step is userA and it is signed number. |

## 4.48 OCC_PACK_TOS_DELTA Register

Address Offset: 0x42

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| occ_pack_tos_delta | [15:0] | 0x14 | RW | This parameter configures an alternate recovery mechanism for Overcurrent in Charge. If the PACK pin voltage is less than or equal to the Top-of-Stack voltage minus this delta for RECOVER_DELAY second, recovery occurs. |

## 4.49 OCD1_THRESHOLD Register

Address Offset: 0x44

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| ocd1_th | [8:0] | 0x19 | RW | This parameter sets the Overcurrent in Discharge 1st Tier Protection threshold for the sense resistor voltage in units of 2mV with 3mv offset. |

## 4.50 OCD1_DELAY Register

Address Offset: 0x46

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| ocd1_delay | [7:0] | 0x4 | RW | This parameter sets the Cell Overcurrent in DisCharge 1st Tier Protection delay in units of comparator loops(each loop is 3.9ms) with a 2 loops offset. |

## 4.51 OCD2_THRESHOLD Register

Address Offset: 0x48

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| ocd2_th | [8:0] | 0x32 | RW | This parameter sets the Overcurrent in Discharge 2nd Tier Protection threshold for the sense resistor voltage in units of 2mV with 3mv offset. |


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## 4.52 OCD2_DELAY Register

Address Offset: 0x4A

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| ocd2_delay | [7:0] | 0x7 | RW | This parameter sets the Cell Overcurrent in DisCharge 2nd Tier Protection delay in units of comparator loops(each loop is 3.9ms) with a 2 loops offset. |

## 4.53 SCD_THRESHOLD Register

Address Offset: 0x4B

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| scd_th | [7:0] | 0x9 | RW | This parameter sets the Short Circuit in Discharge Protection threshold for the sense resistor voltage.<br>0 : 10 mV<br>1 : 20 mV<br>2 : 40 mV<br>3 : 60 mV<br>4 : 80 mV<br>5 : 100 mV<br>6 : 125 mV<br>7 : 150 mV<br>8 : 175 mV<br>9 : 200 mV<br>10 : 250 mV<br>11 : 300 mV<br>12 : 350 mV<br>13 : 400 mV<br>14 : 450 mV<br>15 : 500 mV |

## 4.54 SCD_DELAY Register

Address Offset: 0x4C

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| scd_delay | [7:0] | 0x2 | RW | This parameter sets the delay before the fault is triggered in units of 31.25 µs.<br>1: Protection is enabled with no delay.<br>All other values:Enabled with a delay of (value - 1) * 31.25 µs |


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## 4.55 SCD_RECOVERY_TIME Register

Address Offset: 0x4D

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| scd_recov_time | [7:0] | 0x5 | RW | This parameter configures the delay after which the Short Circuit in Discharge Protection recovers. |

## 4.56 OCD3_THRESHOLD Register

Address Offset: 0x4E

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| ocd3_th | [15:0] | 0xf060 | RW | This parameter sets the Overcurrent in Discharge 3rd Tier Protection threshold. Units of reported current are configurable. This parameter uses the same units as reported current, so it is in units of user-amps. |

## 4.57 OCD3_DELAY Register

Address Offset: 0x50

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| ocd3_delay | [7:0] | 0x2 | RW | This parameter sets the delay for Overcurrent in Discharge 3rd Tier to trigger in units of seconds. |

## 4.58 OCD_RECOVERY_THRESHOLD Register

Address Offset: 0x52

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| ocd_recov_th | [15:0] | 0xc8 | RW | This sets the recovery threshold for Overcurrent in Discharge 1st, 2nd, and 3rd Tier Protections. Measured current must be greater than or equal to this threshold for RECOVER_DELAY to recover. The step is useA and it is signed number. |

## 4.59 OCDL_LATCH_LIMIT Register

Address Offset: 0x54

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| ocdl_latch_limit | [7:0] | 0x2 | RW | Whenever an OCD event occurs, the OCD latch counter is incremented. If the latch counter reaches this limit, the Overcurrent Latch Protection is triggered. This protection allows for alternate recovery conditions when repeated Overcurrent in Discharge events occur. The value of the parameter ocdl_dec_delay must be more than ocdl_recov_delay. |

## 4.60 OCDL_COUNTER_DEC_DELAY Register

Address Offset: 0x55


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| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| ocdl_dec_delay | [7:0] | 0xa | RW | Whenever an OCD event occurs, the OCD latch counter is incremented. While the counter is nonzero, it will be decremented after the delay set by this parameter. |

## 4.61 OCDL_RECOVERY_TIME Register

Address Offset: 0x56

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| ocdl_recov_delay | [7:0] | 0xf | RW | This parameter sets the recovery delay for the Overcurrent in Discharge Latch Protection. Measured current must be greater than or equal to the OCDL_RECOV_TH for this many seconds for recovery to occur via this mechanism. This recovery mechanism is disabled when OCDL_CURR_RECOV is not set.The unit is S. |

## 4.62 OCDL_RECOVERY_THRESHOLD Register

Address Offset: 0x58

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| ocdl_recov_th | [15:0] | 0xc8 | RW | This parameter sets the recovery threshold for the Overcurrent in Discharge Latch Protection..The unit is S. |

## 4.63 SCDL_LATCH_LIMIT Register

Address Offset: 0x5A

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| scdl_latch_limit | [7:0] | 0x2 | RW | Whenever an SCD event occurs, the SCD latch counter is incremented. If the latch counter reaches this limit, the Short Circuit in Discharge Latch Protection is triggered. This protection allows for alternate recovery conditions when repeated Short Circuit in Discharge events occur. |

## 4.64 SCDL_COUNTER_DEC_DELAY Register

Address Offset: 0x5B

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| scdl_dec_delay | [7:0] | 0xa | RW | Whenever an SCD event occurs, the SCD latch counter is incremented. While the counter is nonzero, it will be decremented after the delay set by this parameter. The value of the parameter scdl_dec_delay must be more than scdl_recov_delay.The unit is S. |

## 4.65 SCDL_RECOVERY_TIME Register

Address Offset: 0x5C

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|


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| | | | | |
|---|---|---|---|---|
| scdl_recov_delay | [7:0] | 0xf | RW | This parameter sets the recovery delay for the Short Circuit in Discharge Latch Protection. Measured current must be greater than or equal to the SCDL_RECOV_TH for this many seconds for recovery to occur via this mechanism. This recovery mechanism is disabled when SCDL_CURR_RECOV is not set.The unit is S. |

## 4.66 SCDL_RECOVERY_THRESHOLD Register

Address Offset: 0x5E

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| scdl_recov_th | [15:0] | 0xc8 | RW | his parameter sets the recovery threshold for the Short Circuit in Discharge Latch Protection. |

## 4.67 OTC_THRESHOLD Register

Address Offset: 0x60

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| otc_th | [7:0] | 0x37 | RW | This parameter sets the Overtemperature in Charge Protection threshold. When the maximum cell temperature is greater than or equal to this threshold for OTC_DELAY seconds, the protection is triggered. The unit is ℃, the most significant bit is the sign bit. |

## 4.68 OTC_DELAY Register

Address Offset: 0x61

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| otc_delay | [7:0] | 0x2 | RW | This parameter sets the Overtemperature in Charge Protection delay. When the maximum cell temperature is greater than or equal to OTC_TH for this many seconds, the protection is triggered. |

## 4.69 OTC_RECOVERY_THRESHOLD Register

Address Offset: 0x62

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| otc_recov_th | [7:0] | 0x32 | RW | This parameter sets the Overtemperature in Charge Protection recovery threshold. Recovery occurs when the maximum cell temperature is less than or equal to this threshold for RECOVER_DELAY seconds.the most significant bit is the sign bit. |

## 4.70 OTD_THRESHOLD Register

Address Offset: 0x63

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|


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| | | | | |
|---|---|---|---|---|
| otd_th | [7:0] | 0x3c | RW | This parameter sets the Overtemperature in Discharge Protection threshold. The unit is ℃, the most significant bit is the sign bit. |

## 4.71 OTD_DELAY Register

Address Offset: 0x64

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| otd_delay | [7:0] | 0x2 | RW | This parameter sets the Overtemperature in Discharge Protection delay. |

## 4.72 OTD_RECOVERY_THRESHOLD Register

Address Offset: 0x65

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| otd_recov_th | [7:0] | 0x37 | RW | This parameter sets the Overtemperature in Discharge Protection recovery threshold. The unit is ℃, the most significant bit is the sign bit. |

## 4.73 OTF_THRESHOLD Register

Address Offset: 0x66

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| otf_th | [7:0] | 0x50 | RW | This parameter sets the FET Overtemperature threshold. The unit is ℃, the most significant bit is the sign bit. |

## 4.74 OTF_DELAY Register

Address Offset: 0x67

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| otf_delay | [7:0] | 0x2 | RW | This parameter sets the FET Overtemperature Protection delay. |

## 4.75 OTF_RECOVERY_THRESHOLD Register

Address Offset: 0x68

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| otf_recov_th | [7:0] | 0x41 | RW | This parameter sets the FET Overtemperature Protection recovery threshold. . The unit is ℃, the most significant bit is the sign bit. |

## 4.76 OTINT_THRESHOLD Register

Address Offset: 0x69

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| otint_th | [7:0] | 0x55 | RW | This parameter sets the Internal Overtemperature threshold. The unit is ℃, the most significant bit is the sign bit. |


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## 4.77 OTINT_DELAY Register

Address Offset: 0x6A

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| otint_delay | [7:0] | 0x2 | RW | This parameter sets the Internal Overtemperature Protection delay. |

## 4.78 OTINT_RECOVERY_THRESHOLD Register

Address Offset: 0x6B

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| otint_recov_th | [7:0] | 0x50 | RW | This parameter sets the Internal Overtemperature Protection recovery threshold. . The unit is ℃, the most significant bit is the sign bit. |

## 4.79 UTC_THRESHOLD Register

Address Offset: 0x6C

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| utc_th | [7:0] | 0x0 | RW | This parameter sets the Undertemperature in Charge Protection threshold. The unit is ℃, the most significant bit is the sign bit. |

## 4.80 UTC_DELAY Register

Address Offset: 0x6D

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| utc_delay | [7:0] | 0x2 | RW | This parameter sets the Undertemperature in Charge Protection delay. |

## 4.81 UTC_RECOVERY_THRESHOLD Register

Address Offset: 0x6E

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| utc_recov_th | [7:0] | 0x5 | RW | This parameter sets the Undertemperature in Charge Protection recovery threshold. The unit is ℃, the most significant bit is the sign bit. |

## 4.82 UTD_THRESHOLD Register

Address Offset: 0x6F

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| utd_th | [7:0] | 0x0 | RW | This parameter sets the Undertemperature in Discharge Protection threshold. The unit is ℃, the most significant bit is the sign bit. |

## 4.83 UTD_DELAY Register


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Address Offset: 0x70

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| utd_delay | [7:0] | 0x2 | RW | This parameter sets the Undertemperature in Discharge Protection delay. |

## 4.84 UTD_RECOVERY_THRESHOLD Register

Address Offset: 0x71

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| utd_recov_th | [7:0] | 0x5 | RW | This parameter sets the Undertemperature in Discharge Protection recovery threshold.The unit is ℃, the most significant bit is the sign bit. |

## 4.85 UTINT_THRESHOLD Register

Address Offset: 0x72

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| utint_th | [7:0] | 0xec | RW | This parameter sets the Internal Undertemperature threshold. The unit is ℃, the most significant bit is the sign bit. |

## 4.86 UTINT_DELAY Register

Address Offset: 0x73

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| utint_delay | [7:0] | 0x2 | RW | This parameter sets the Internal Undertemperature Protection delay. |

## 4.87 UTINT_RECOVERY_THRESHOLD Register

Address Offset: 0x74

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| utint_recov_th | [7:0] | 0xf1 | RW | This parameter sets the Internal Undertemperature Protection recovery threshold. The unit is °C, the most significant bit is the sign bit. |

## 4.88 RECOVERY_TIME Register

Address Offset: 0x75

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| recov_delay | [7:0] | 0x3 | RW | This parameter configures the recovery time used by several protections in units of seconds. |

## 4.89 HWD_DELAY Register

Address Offset: 0x76

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|


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| | | | | |
|---|---|---|---|---|
| hwd_delay | [15:0] | 0x3c | RW | This parameter configures the Host Watchdog timeout. If communications are not received for this many seconds, the Host Watchdog Fault is triggered. |

## 4.90 PTO_CHARGE_THRESHOLD Register

Address Offset: 0x78

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| pto_chg_th | [15:0] | 0xfa | RW | This parameter sets the current threshold for the Precharge Timeout Protection. When in PRECHARGE mode but the current is less than or equal to this threshold, the timeout timer is not incremented and the Precharge Timeout Suspend (PTOS) bit is set. |

## 4.91 PTO_DELAY Register

Address Offset: 0x7A

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| pto_delay | [15:0] | 0x708 | RW | This parameter sets the Precharge Timeout threshold. When in PRECHARGE mode with current above PTO_CHG_TH, a timer is incremented. If the timer reaches this value, the Precharge Timeout Protection is triggered. |

## 4.92 PTO_RESET Register

Address Offset: 0x7C

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| pto_reset | [15:0] | 0x2 | RW | The timer for the Precharge Timeout Protection is reset when a continuous discharge of this much capacity occurs. This discharge must be larger in magnitude than DSG_CURRENT_TH for long enough to accumulate this amount of charge. |

## 4.93 CUDEP_THRESHOLD Register

Address Offset: 0x7E

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cudep_th | [15:0] | 0x5dc | RW | When CUDEP PF is enabled, the copper deposition check holds the FETs off until the voltage is greater than or equal to CUDEP_TH for CUDEP_DELAY seconds. If the voltage is below CUDEP_TH for CUDEP_DELAY seconds, the CUDEP Permanent Fail is triggered. |

## 4.94 CUDEP_DELAY Register

Address Offset: 0x80

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|


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| | | | | |
|---|---|---|---|---|
| cudep_delay | [7:0] | 0x2 | RW | When CUDEP PF is enabled, the copper deposition check holds the FETs off until the voltage is greater than or equal to CUDEP_TH for CUDEP_DELAY seconds. If the voltage is below CUDEP_TH for CUDEP_DELAY seconds, the CUDEP Permanent Fail is triggered. |

## 4.95 SUV_THRESHOLD Register

Address Offset: 0x82

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| suv_th | [15:0] | 0x898 | RW | This parameter sets the Safety Undervoltage Permanent Fail threshold. When the minimum cell voltage is less than or equal to SUV_TH for SUV_DELAY seconds, the SUV Permanent Fail is triggered. |

## 4.96 SUV_DELAY Register

Address Offset: 0x84

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| suv_delay | [7:0] | 0x5 | RW | This parameter sets the Safety Undervoltage Permanent Fail threshold. When the minimum cell voltage is less than or equal to SUV_TH for SUV_DELAY seconds, the SUV Permanent Fail is triggered. |

## 4.97 SOV_THRESHOLD Register

Address Offset: 0x86

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| sov_th | [15:0] | 0x1194 | RW | This parameter sets the Safety Undervoltage Permanent Fail threshold. When the maximum cell voltage is greater than or equal to SOV_TH for SOV_DELAY seconds, the SOV Permanent Fail is triggered. |

## 4.98 SOV_DELAY Register

Address Offset: 0x88

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| sov_delay | [7:0] | 0x5 | RW | This parameter sets the Safety Undervoltage Permanent Fail threshold. When the maximum cell voltage is greater than or equal to SOV_TH for SOV_DELAY seconds, the SOV Permanent Fail is triggered. |


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## 4.99 TOS_THRESHOLD Register

Address Offset: 0x8A

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| tos_th | [15:0] | 0x1f4 | RW | This parameter sets the Top of Stack versus Cell Sum Permanent Fail threshold. The threshold is based on cell voltage, so it is multiplied by the number of used cells before the comparison. The device compares the measured Top of Stack voltage to the sum of individually measured cell voltages. When the absolute value of the difference is greater than or equal to the number of used cells times the TOS_TH for TOS_DELAY seconds, the TOSF Permanent Fail is triggered. |

## 4.100 TOS_DELAY Register

Address Offset: 0x8C

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| tos_delay | [7:0] | 0x5 | RW | This parameter sets the Top of Stack versus Cell Sum Permanent Fail threshold. The threshold is based on cell voltage, so it is multiplied by the number of used cells before the comparison. The device compares the measured Top of Stack voltage to the sum of individually measured cell voltages. When the absolute value of the difference is greater than or equal to the number of used cells times the TOS_TH for TOS_DELAY seconds, the TOSF Permanent Fail is triggered. |

## 4.101 SOCC_THRESHOLD Register

Address Offset: 0x8E

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| socc_th | [15:0] | 0x2710 | RW | This parameter sets the Safety Overcurrent in Charge Permanent Fail threshold. When the measured current is greater than or equal to SOCC_TH for SOCC_DELAY seconds, the SOCC Permanent Fail is triggered. |

## 4.102 SOCC_DELAY Register

Address Offset: 0x90

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| socc_delay | [7:0] | 0x5 | RW | This parameter sets the Safety Overcurrent in Charge Permanent Fail threshold. When the measured current is greater than or equal to SOCC_TH for SOCC_DELAY seconds, the SOCC Permanent Fail is triggered. |


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## 4.103 SOCD_THRESHOLD Register

Address Offset: 0x92

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| socd_th | [15:0] | 0x8300 | RW | This parameter sets the Safety Overcurrent in Discharge Permanent Fail threshold. When the measured current is greater than or equal to SOCD_TH for SOCD_DELAY seconds, the SOCD Permanent Fail is triggered. |

## 4.104 SOCD_DELAY Register

Address Offset: 0x94

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| socd_delay | [7:0] | 0x5 | RW | This parameter sets the Safety Overcurrent in Discharge Permanent Fail threshold. When the measured current is greater than or equal to SOCD_TH for SOCD_DELAY seconds, the SOCD Permanent Fail is triggered. |

## 4.105 SOT_THRESHOLD Register

Address Offset: 0x95

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| sot_th | [7:0] | 0x41 | RW | This parameter sets the Safety Overtemperature Permanent Fail threshold. When the maximum cell temperature is greater than or equal to SOT_TH for SOT_DELAY seconds, the SOT Permanent Fail is triggered. |

## 4.106 SOT_DELAY Register

Address Offset: 0x96

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| sot_delay | [7:0] | 0x5 | RW | This parameter sets the Safety Overtemperature Permanent Fail threshold. When the maximum cell temperature is greater than or equal to SOT_TH for SOT_DELAY seconds, the SOT Permanent Fail is triggered. |

## 4.107 SOTF_THRESHOLD Register

Address Offset: 0x97

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| sotf_th | [7:0] | 0x55 | RW | This parameter sets the Safety Overtemperature FET Permanent Fail threshold. When the fet temperature is greater than or equal to SOTF_TH for SOTF_DELAY seconds, the SOTF Permanent Fail is triggered. |


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## 4.108 SOTF_DELAY Register

Address Offset: 0x98

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| sotf_delay | [7:0] | 0x5 | RW | This parameter sets the Safety Overtemperature FET Permanent Fail threshold. When the fet temperature is greater than or equal to SOTF_TH for SOTF_DELAY seconds, the SOTF Permanent Fail is triggered. |

## 4.109 VIMR_CHECK_VOLTAGE Register

Address Offset: 0x9A

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| vimr_chk_voltage | [15:0] | 0xdac | RW | The Voltage Imbalance at Rest Permanent Fail is not checked when the maximum cell voltage is less than this threshold. This is because at lower states of charge, the voltage differences for a smaller imbalance are amplified. |

## 4.110 VIMR_MAX_RELAX_CURRENT Register

Address Offset: 0x9C

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| vimr_max_relax_curr | [15:0] | 0xa | RW | The Voltage Imbalance at Rest Permanent Fail is not checked when the absolute value of measured current is greater than or equal to this threshold. |

## 4.111 VIMR_THRESHOLD Register

Address Offset: 0x9E

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| vimr_th | [15:0] | 0x1f4 | RW | This parameter sets the Voltage Imbalance at Rest Permanent Fail threshold. When the delta between the maximum and minimum cell voltages is greater than or equal to VIMR_TH for VIMR_DELAY seconds while VIMR check conditions are met, the VIMR Permanent Fail is triggered. |

## 4.112 VIMR_DELAY Register

Address Offset: 0xA0

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| vimr_delay | [7:0] | 0x5 | RW | This parameter sets the Voltage Imbalance at Rest Permanent Fail threshold. When the delta between the maximum and minimum cell voltages is greater than or equal to VIMR_TH for VIMR_DELAY seconds while VIMR check conditions are met, the VIMR Permanent Fail is triggered. |


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## 4.113 VIMR_RELAX_MIN_DURATION Register

Address Offset: 0xA2

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| vimr_relax_min_dura | [15:0] | 0x64 | RW | The Voltage Imbalance at Rest Permanent Fail is not checked unless the VIMR_CHK_VOLTAGE and VIMR_MAX_RELAX_CURR conditions have been met for this duration. |

## 4.114 VIMA_CHECK_VOLTAGE Register

Address Offset: 0xA4

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| vima_chk_voltage | [15:0] | 0xe74 | RW | The Voltage Imbalance Active Permanent Fail is not checked when the maximum cell voltage is less than this threshold. This is because at lower states of charge, the voltage differences for a smaller imbalance are amplified. |

## 4.115 VIMA_MIN_ACTIVE_CURRENT Register

Address Offset: 0xA6

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| vima_min_active_curr | [15:0] | 0x32 | RW | The Voltage Imbalance Active Permanent Fail is not checked when the absolute value of measured current is less than this threshold. |

## 4.116 VIMA_THRESHOLD Register

Address Offset: 0xA8

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| vima_th | [15:0] | 0xc8 | RW | This parameter sets the Voltage Imbalance Active Permanent Fail threshold. When the delta between the maximum and minimum cell voltages is greater than or equal to VIMA_TH for VIMA_DELAY seconds while VIMA check conditions are met, the VIMA Permanent Fail is triggered. |

## 4.117 VIMA_DELAY Register

Address Offset: 0xAA

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| vima_delay | [7:0] | 0x5 | RW | This parameter sets the Voltage Imbalance Active Permanent Fail threshold. When the delta between the maximum and minimum cell voltages is greater than or equal to VIMA_TH for VIMA_DELAY seconds while VIMA check conditions are met, the VIMA Permanent Fail is triggered. |


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## 4.118 CFETF_OFF_THRESHOLD Register

Address Offset: 0xAC

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cfetf_off_th | [15:0] | 0x14 | RW | This parameter sets the Charge FET Permanent Fail current threshold. When the measured current is greater than or equal to CFETF_OFF_TH for CFETF_OFF_DELAY seconds while the CHG FET is off, the CFETF Permanent Fail is triggered. |

## 4.119 CFETF_OFF_DELAY Register

Address Offset: 0xAE

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cfetf_off_delay | [7:0] | 0x5 | RW | This parameter sets the Charge FET Permanent Fail current threshold. When the measured current is greater than or equal to CFETF_OFF_TH for CFETF_OFF_DELAY seconds while the CHG FET is off, the CFETF Permanent Fail is triggered. |

## 4.120 DFETF_OFF_THRESHOLD Register

Address Offset: 0xB0

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| dfetf_off_th | [15:0] | 0xffec | RW | This parameter sets the Discharge FET Permanent Fail current threshold. When the measured current is less (more negative) than or equal to DFETF_OFF_TH for DFETF_OFF_DELAY seconds while the DSG FET is off, the DFETF Permanent Fail is triggered. |

## 4.121 DFETF_OFF_DELAY Register

Address Offset: 0xB2

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| dfetf_off_delay | [7:0] | 0x5 | RW | This parameter sets the Discharge FET Permanent Fail current threshold. When the measured current is less (more negative) than or equal to DFETF_OFF_TH for DFETF_OFF_DELAY seconds while the DSG FET is off, the DFETF Permanent Fail is triggered. |


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## 4.122 VSSF_FAIL_THRESHOLD Register

Address Offset: 0xB4

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| vssf_fail_th | [15:0] | 0x64 | RW | This parameter sets the Internal VSS Measurement Permanent Fail threshold. When the ADC count value measured for the internal VSS channel is greater than or equal to VSSF_FAIL_TH for VSSF_DELAY seconds, the VSSF Permanent Fail is triggered. |

## 4.123 VSSF_DELAY Register

Address Offset: 0xB6

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| vssf_delay | [7:0] | 0x5 | RW | This parameter sets the Internal VSS Measurement Permanent Fail threshold. When the ADC count value measured for the internal VSS channel is greater than or equal to VSSF_FAIL_TH for VSSF_DELAY seconds, the VSSF Permanent Fail is triggered. |

## 4.124 DUALLVL_DELAY Register

Address Offset: 0xB7

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| dlvl_delay | [7:0] | 0x5 | RW | This parameter sets the Second Level Protector Permanent Fail delay. The device checks the FUSE pin each second, and if it is asserted by the second level protector for DLVL_DELAY seconds, the 2LVL PF is triggered. |

## 4.125 LFOF_DELAY Register

Address Offset: 0xB8

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| lfof_delay | [7:0] | 0x5 | RW | This parameter sets the Internal LFO Permanent Fail delay. The device monitors the output of its Low-Frequency Oscillator integrity check. If the error bit is set for LFOF_DELAY seconds, the LFOF Permanent Failure is triggered. |

## 4.126 HWMX_DELAY Register

Address Offset: 0xBA

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| hwmx_delay | [7:0] | 0x5 | RW | This parameter sets the Hardware Mux Permanent Fail delay. The device periodically performs an integrity check of the input mux for the hardware protection comparator subsystem, which is used for the OV, UV, OCC, OCD1, and |


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| | | | | |
|---|---|---|---|---|
| | | | | OCD2 primary protections. If this check fails for HWMX_DELAY seconds, the Hardware Mux Permanent Failure is triggered. |

## 4.127 SF_FAULT_MASK_A Register

Address Offset: 0xBB

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| scd_sf_fault_mask | [7:7] | 0x1 | RW | When the bitwise of this bit and SCD SafetyFault is nonzero AlarmRawStatus()[MSK_SFFault] is set. |
| ocd2_sf_fault_mask | [6:6] | 0x1 | RW | When the bitwise of this bit and OCD2 SafetyFault is nonzero AlarmRawStatus()[MSK_SFFault] is set. |
| ocd1_sf_fault_mask | [5:5] | 0x1 | RW | When the bitwise of this bit and OCD1 SafetyFault is nonzero AlarmRawStatus()[MSK_SFFault] is set. |
| occ_sf_fault_mask | [4:4] | 0x1 | RW | When the bitwise of this bit and OCC SafetyFault is nonzero AlarmRawStatus()[MSK_SFFault] is set. |
| cov_sf_fault_mask | [3:3] | 0x1 | RW | When the bitwise of this bit and COV SafetyFault is nonzero AlarmRawStatus()[MSK_SFFault] is set. |
| cuv_sf_fault_mask | [2:2] | 0x1 | RW | When the bitwise of this bit and CUV SafetyFault is nonzero AlarmRawStatus()[MSK_SFFault] is set. |

## 4.128 SF_FAULT_MASK_B Register

Address Offset: 0xBC

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| otf_sf_fault_mask | [7:7] | 0x1 | RW | When the bitwise of this bit and OTF SafetyFault is nonzero AlarmRawStatus()[MSK_SFFault] is set. |
| otint_sf_fault_mask | [6:6] | 0x1 | RW | When the bitwise of this bit and OTINT SafetyFault is nonzero AlarmRawStatus()[MSK_SFFault] is set. |
| otd_sf_fault_mask | [5:5] | 0x1 | RW | When the bitwise of this bit and OTD SafetyFault is nonzero AlarmRawStatus()[MSK_SFFault] is set. |
| otc_sf_fault_mask | [4:4] | 0x1 | RW | When the bitwise of this bit and OTC SafetyFault is nonzero AlarmRawStatus()[MSK_SFFault] is set. |
| utint_sf_fault_mask | [2:2] | 0x1 | RW | When the bitwise of this bit and UTINT SafetyFault is nonzero AlarmRawStatus()[MSK_SFFault] is set. |
| utd_sf_fault_mask | [1:1] | 0x1 | RW | When the bitwise of this bit and UTD SafetyFault is nonzero AlarmRawStatus()[MSK_SFFault] is set. |
| utc_sf_fault_mask | [0:0] | 0x1 | RW | When the bitwise of this bit and UTC SafetyFault is nonzero AlarmRawStatus()[MSK_SFFault] is set. |


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## 4.129 SF_FAULT_MASK_C Register

Address Offset: 0xBD

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| ocd3_sf_fault_mask | [7:7] | 0x1 | RW | When the bitwise of this bit and OCD3 SafetyFault is nonzero AlarmRawStatus()[MSK_SFFault] is set. |
| scdl_sf_fault_mask | [6:6] | 0x1 | RW | When the bitwise of this bit and SCDL SafetyFault is nonzero AlarmRawStatus()[MSK_SFFault] is set. |
| ocdl_sf_fault_mask | [5:5] | 0x1 | RW | When the bitwise of this bit and OCDL SafetyFault is nonzero AlarmRawStatus()[MSK_SFFault] is set. |
| covl_sf_fault_mask | [4:4] | 0x1 | RW | When the bitwise of this bit and COVL SafetyFault is nonzero AlarmRawStatus()[MSK_SFFault] is set. |
| pto_sf_fault_mask | [2:2] | 0x1 | RW | When the bitwise of this bit and PTO SafetyFault is nonzero AlarmRawStatus()[MSK_SFFault] is set. |
| hwd_sf_fault_mask | [1:1] | 0x1 | RW | When the bitwise of this bit and HWD SafetyFault is nonzero AlarmRawStatus()[MSK_SFFault] is set. |


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# 5 Cb

Base Address: 0x8C00

## 5.1 BALANCING_CONFIGURATION Register

Address Offset: 0x00

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| opw_en | [7:7] | 0x1 | RW | 0: open_wire detection function is disabled<br>1: open_wire detection function is enabled |
| cb_en | [6:6] | 0x0 | RW | 0: cell balancing function is disabled<br>1: cell balancing function is enabled |
| cb_alert_mask | [5:5] | 0x0 | RW | 0: ALERT Dont Break cell balance<br>1: ALERT Will Break cell balance |
| cb_no_cmd | [4:4] | 0x0 | RW | 0: Host-controlled balancing commands are accepted<br>1: Host-controlled balancing commands are ignored |
| cb_nosleep | [3:3] | 0x0 | RW | 0: sleep mode is allowed while cell balancing is active<br>1: sleep mode is prevented while cell balancing is active |
| cb_sleep | [2:2] | 0x0 | RW | 0: Cell balancing is not performed in sleep mode<br>1: Cell balancing may be performed in sleep mode |
| cb_rlx | [1:1] | 0x0 | RW | 0: Cell balancing is not allowed in relax conditions<br>1: Cell balancing is allowed in relax conditions |
| cb_chg | [0:0] | 0x0 | RW | 0: Cell balancing is not allowed while charging<br>1: Cell balancing is allowed while charging |

## 5.2 CB_MIN_CELL_TEMP Register

Address Offset: 0x01

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| min_cell_temp | [7:0] | 0xec | RW | When the minimum cell temperature is below this value, cell balancing is not allowed. This affects both host-controlled balancing and automatic cell balancing Configure Range in -128~127. |

## 5.3 CB_MAX_CELL_TEMP Register

Address Offset: 0x02

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| max_cell_temp | [7:0] | 0x3c | RW | When the maximum cell temperature is above this value, cell balancing is not allowed. This affects both host-controlled balancing and automatic cell balancing Configure Range in -128~127. |


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## 5.4 CB_MAX_INTERNAL_TEMP Register

Address Offset: 0x03

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| max_int_temp | [7:0] | 0x46 | RW | When the internal temperature is above this value, cell balancing is not allowed. This affects both host-controlled balancing and automatic cell balancing Configure Range in -128~127. |

## 5.5 CB_BALANCE_INTERVAL Register

Address Offset: 0x04

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cb_interval | [7:0] | 0x14 | RW | The cell balancing algorithm recalculates which cell to balance after this many seconds. Once a cell is chosen, balancing on that cell will continue for this interval unless one of the conditions which blocks balancing is present. This interval is also used with the commands for host-controlled balancing. A command to balance a cell will keep balancing active for this amount of time. The host may send the command again before the timer expires to reset it and continue balancing. |

## 5.6 CB_BALANCE_MAX_CELLS Register

Address Offset: 0x05

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cb_max_cells | [7:0] | 0x1 | RW | This limits how many cells may be automatically balanced in parallel or balanced via the CB_SET_LVL command. With host-controlled balancing, the host may specify more cells than this and the limit is ignored. |

## 5.7 CB_MIN_CELL_V_CHARGE Register

Address Offset: 0x06

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cb_min_volt_chg | [15:0] | 0xf3c | RW | While charging, automatic cell balancing is disabled if the minimum cell voltage is less than this threshold. |

## 5.8 CB_MIN_DELTA_CHARGE Register

Address Offset: 0x08

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cb_min_delta_chg | [7:0] | 0x28 | RW | While charging, the delta between the maximum and minimum cell voltages must be greater than this value for automatic cell balancing to begin. |


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## 5.9 CB_STOP_DELTA_CHARGE Register

Address Offset: 0x09

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cb_stop_delta_chg | [7:0] | 0x14 | RW | While balancing in charge, this value is used as the target voltage delta to balance to. Once the delta between the minimum and maximum cell voltages falls below this threshold, balancing stops. This provides some hysteresis around the delta threshold. Note that to prevent balancing from draining the pack, the stop threshold is calculated when balancing begins and will not decrease if the minimum cell voltage drops. This means that in some conditions, balancing will stop will a greater delta than this. |

## 5.10 CB_MIN_CELL_V_RELAX Register

Address Offset: 0x0A

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cb_min_volt_rlx | [15:0] | 0xf3c | RW | While not charging or discharging, automatic cell balancing is disabled if the minimum cell voltage is less than this threshold. |

## 5.11 CB_MIN_DELTA_RELAX Register

Address Offset: 0x0C

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cb_min_delta_rlx | [7:0] | 0x28 | RW | While not charging or discharging, the delta between the maximum and minimum cell voltages must be greater than this value for automatic cell balancing to begin. |

## 5.12 CB_STOP_DELTA_RELAX Register

Address Offset: 0x0D

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cb_stop_delta_rlx | [7:0] | 0x14 | RW | While balancing in relax, this value is used as the target voltage delta to balance to. Once the delta between the minimum and maximum cell voltages falls below this threshold, balancing stops. This provides some hysteresis around the delta threshold. Note that to prevent balancing from draining the pack, the stop threshold is calculated when balancing begins and will not decrease if the minimum cell voltage drops. This means that in some conditions, balancing will stop will a greater delta than this. |


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## 5.13 LD_ACTIVE_TIME Register

Address Offset: 0x0E

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| ld_active_time | [7:0] | 0x0 | RW | The Short Circuit in Discharge Latch and Overcurrent in Discharge Latch protections can be configured to recover when the load removal is detected. The device can enable a current source on the LD pin and checks for the voltage level rising above ld_remove_voltage. This parameter controls how long the current source is enabled before waiting LD_RETRY_DELAY seconds and trying again. This should be set longer than Sleep_Voltage_Time to ensure that voltage is measured while the current source is enabled.<br>0:Recovery based on Load Detect functionality is disabled.<br>All other values:Load Detect current source is enabled for this many seconds at a time. |

## 5.14 LD_RETRY_DELAY Register

Address Offset: 0x0F

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| ld_retry_delay | [7:0] | 0x32 | RW | The Short Circuit in Discharge Latch and Overcurrent in Discharge Latch protections can be configured to recover when the load removal is detected. The device can enable a current source on the LD pin and checks for the voltage level rising above ld_remove_voltage. The current source is enabled only temporarily to save power. This parameter controls the delay between periods when the LD current source is active.<br>0:Load Detect current source stays on until LD_TIMEOUT or recovery.<br>All other values:Load detect current source is disabled for this many seconds before trying again. |

## 5.15 LD_TIMEOUT Register

Address Offset: 0x10

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| ld_timeout | [15:0] | 0x1 | RW | To prevent the Load Detect function from indefinitely operating and draining the battery, a timeout may be configured. This timeout starts when Load Detect is first enabled to check recovery and is measured in hours. After this timeout, Load Detect is no longer checked until the latch faults have recovered or a command to retry is received. |


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## 5.16 DSG_CURRENT_THRESHOLD Register

Address Offset: 0x12

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| dsg_current_th | [15:0] | 0x64 | RW | Certain device features depend upon whether the system is in a discharge state or not. The system is considered to be discharging when measured negative current is larger in magnitude than this threshold. The unit is userA, userA is user-defined in cc gain |

## 5.17 CHG_CURRENT_THRESHOLD Register

Address Offset: 0x14

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| chg_current_th | [15:0] | 0x32 | RW | Certain device features depend upon whether the system is in a charging state or not. The system is considered to be charging when measured positive current is larger in magnitude than this threshold. The unit is userA, userA is user-defined in cc gain |

## 5.18 COW_CHECK_TIME Register

Address Offset: 0x16

| Field | Bits | Default | Type | Field Description |
|---|---|---|---|---|
| cow_check_time | [7:0] | 0x5 | RW | If a cell connection to the device is detached, the capacitance on the node may maintain the voltage for some time. To detect this condition, the device can periodically enable a current source to VSS on each cell connection to drain the charge on the node. When a cell is detached, that node voltage is discharged, and an over-voltage or under-voltage event will be triggered on that cell or an adjacent cell differential measurement. The current source is enabled for the duration of an ADC measurement once per the interval defined by this parameter. A longer interval decreases the average current consumption at the cost of detection time. When using this feature, cell balancing should be considered as well since the current flows from the cell pin to VSS, resulting in a higher average current drawn from lower cells.<br>0:Cell open-wire check is disabled.<br>All other values:Cell open-wire check current source is enabled briefly for each cell on this interval. |


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# 6 Revision History

| REVISION NUMBER | DESCRIPTION | DATE |
|---|---|---|
| A | Initial release | Apr 29, 2025 |
| B | 1. Modify the description of dchg_pin_fxn、ddsg_pin_fxn、ts0_opt、pin_mode、busbar_sel description.<br>2. Deleted TS4-TS9 registers.<br>3. Added vcell_mode description. | May 22, 2025 |
| C | Update the unit | Dec 15, 2025 |
| D | Update register description: cc2 avg sample、shutdown temperature、auto shutdown time、spi config | Mar 20, 2026 |
| E | Update OCC/ODC1/OCD2 description, Update DA config description, Update comm idle description. Update OCDL&SCDL dec_count description. | Apr 15, 2026 |


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